

Combinatorial Mathematics



Chapter 9

Relations



Outline

- 9.1 Relations and their properties
- 9.3 Representing Relations
- 9.4 Closures of Relations
- 9.5 Equivalence Relations
- 9.6 Partial Orderings



9.1 Relations and Their properties.

✕The most direct way to express a relationship between elements of two sets is to use ordered pairs.

For this reason, sets of ordered pairs are called **binary relations**.

Def 1 $R \subseteq A \times B$.

Let A and B be sets. A **binary relation from A to B** is a subset R of $A \times B = \{ (a, b) : a \in A, b \in B \}$.

Example 1.

A : the set of students in your school.

B : the set of courses.

$R = \{ (a, b) : a \in A, b \in B, a \text{ is enrolled in course } b \}$

Def 1'. We use the notation aRb to denote that $(a, b) \in R$, and $a \not R b$ to denote that $(a, b) \notin R$.

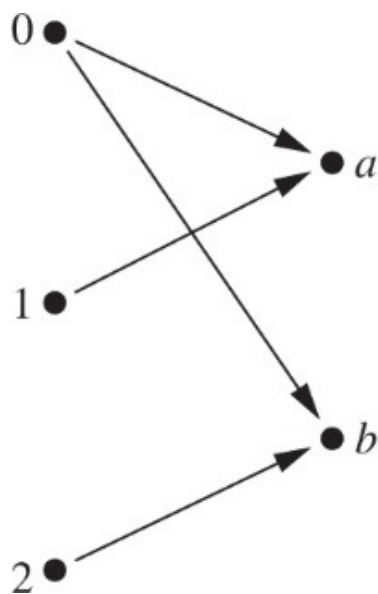
Moreover, a is said to be related to b by R if aRb .

Example 3. Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$, then $\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation R from A to B .

This means, for instance, that $0Ra$, but that $1 \not R b$.

$$R \subseteq A \times B = \{ (0, a), (0, b), (1, a), (1, b), (2, a), (2, b) \}$$

$\notin R \quad \notin R$



R	a	b
0	×	×
1	×	
2		×

- Relations are more general than functions. A function is a relation where exactly one element of B is related to each element of A .



Examples: A : 男生, B : 女生, R : 夫妻關係
 A : 城市, B : 州/省, R : 屬於 (Example 2)

Note. Relations vs. Functions

A relation can be used to express a 1-to-many relationship between the elements of the sets A and B .

(function 不可一對多，只可多對一)

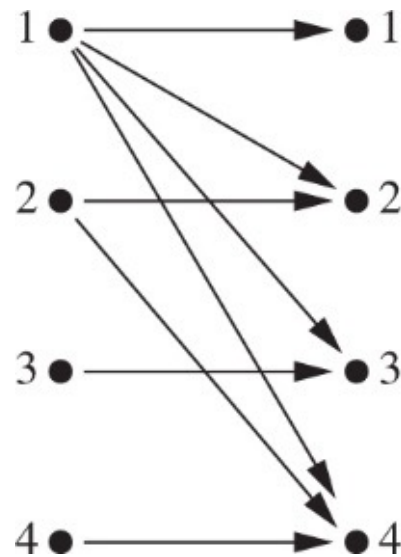
Properties on a Set

Def 2. A relation R on the set A is a relation from A to A . (i.e., R is a subset of $A \times A$)

Example 4.

Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{ (a, b) \mid a \text{ divides } b \}$?

Sol :



R	1	2	3	4
1	×	×	×	×
2		×		×
3			×	
4				×

$$R = \{ (1,1), (1,2), (1,3), (1,4), \\ (2,2), (2,4), \\ (3,3), \\ (4,4) \}$$

Example 5. Consider the following relations on $\mathbb{Z}$.

$$R_1 = \{ (a, b) \mid a \leq b \}$$

$$R_2 = \{ (a, b) \mid a > b \}$$

$$R_3 = \{ (a, b) \mid a = b \text{ or } a = -b \}$$

$$R_4 = \{ (a, b) \mid a = b \}$$

$$R_5 = \{ (a, b) \mid a = b+1 \}$$

$$R_6 = \{ (a, b) \mid a + b \leq 3 \}$$

Which of these relations contain each of the pairs $(1,1)$, $(1,2)$, $(2,1)$, $(1,-1)$, and $(2,2)$?

Sol :

	$(1,1)$	$(1,2)$	$(2,1)$	$(1,-1)$	$(2,2)$
R_1	●	●			●
R_2			●	●	
R_3	●			●	●
R_4	●				●
R_5			●		
R_6	●	●	●	●	



Example 6. How many relations are there on a set with n elements?

Sol :

A relation on a set A is a subset of $A \times A$.

$\Rightarrow A \times A$ has n^2 elements. (ordered pairs)

$\Rightarrow A \times A$ has 2^{n^2} subsets.

$\Rightarrow$ There are 2^{n^2} relations.

Properties of Relations

Def 3. A relation R on a set A is called reflexive (反身性) if $(a,a) \in R$ for every $a \in A$.

Remark: Using quantifiers we see that the relation R on the set A is reflexive if $\forall a((a, a) \in R)$, where the universe of discourse is the set of all elements in A .

Example 7. Consider the following relations on

$$\{1, 2, 3, 4\} :$$

$$R_2 = \{ (1,1), (1,2), (2,1) \}$$

$$R_3 = \{ (1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4) \}$$

$$R_4 = \{ (2,1), (3,1), (3,2), (4,1), (4,2), (4,3) \}$$

which of them are reflexive ?

Sol :

$$R_3$$

Reflexive Relations

Definition: R is **reflexive** iff $(a,a) \in R$ for every element $a \in A$.
Written symbolically, R is reflexive if and only if

$$\forall x [x \in U \rightarrow (x, x) \in R]$$

Example 8: The following relations on the integers are reflexive:

$$R_1 = \{(a, b) \mid a \leq b\},$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a, b) \mid a = b\}.$$

If $A = \emptyset$ then the empty relation is reflexive vacuously. That is the empty relation on an empty set is reflexive!

The following relations are not reflexive:

$$R_2 = \{(a, b) \mid a > b\} \quad (\text{note that } 3 \not> 3),$$

$$R_5 = \{(a, b) \mid a = b + 1\} \quad (\text{note that } 3 \neq 3 + 1),$$

$$R_6 = \{(a, b) \mid a + b \leq 3\} \quad (\text{note that } 4 + 4 \not\leq 3).$$



Example 8. Which of the relations from Example 5 are reflexive?

$$R_1 = \{ (a, b) \mid a \leq b \}$$

$$R_2 = \{ (a, b) \mid a > b \}$$

$$R_3 = \{ (a, b) \mid a = b \text{ or } a = -b \}$$

$$R_4 = \{ (a, b) \mid a = b \}$$

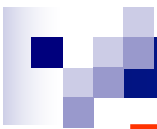
$$R_5 = \{ (a, b) \mid a = b+1 \}$$

$$R_6 = \{ (a, b) \mid a + b \leq 3 \}$$

Sol : R_1, R_3 and R_4

Example 9. Is the “divides” relation on the set of positive integers reflexive?

Sol : Yes.



Def 4.

(1) A relation R on a set A is called symmetric

if for $a, b \in A$,

$$(a, b) \in R \Rightarrow (b, a) \in R.$$

(2) A relation R on a set A is called antisymmetric (反對稱) if for $a, b \in A$,

$$(a, b) \in R \text{ and } (b, a) \in R \Rightarrow a = b.$$

即若 $a \neq b$ 且 $(a, b) \in R \Rightarrow (b, a) \notin R$

Remark: Using quantifiers, we see that the relation R on the set A is symmetric if $\forall a \forall b ((a, b) \in R \rightarrow (b, a) \in R)$. Similarly, the relation R on the set A is antisymmetric if $\forall a \forall b (((a, b) \in R \wedge (b, a) \in R) \rightarrow (a = b))$.



Example 10. Which of the relations from Example 7 are symmetric or antisymmetric ?

$$R_2 = \{ (1,1), (1,2), (2,1) \}$$

$$R_3 = \{ (1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4) \}$$

$$R_4 = \{ (2,1), (3,1), (3,2), (4,1), (4,2), (4,3) \}$$

Sol :

R_2, R_3 are symmetric

R_4 are antisymmetric.

Example 11. Is the “divides” relation on the set of **positive integers** symmetric? Is it antisymmetric?

Sol : It is not symmetric since $1|2$ but $2 \nmid 1$.

It is antisymmetric since $a|b$ and $b|a$ implies $a=b$.

■ 補充：

antisymmetric 跟 symmetric可並存

$$\forall (a, b) \in R, a \neq b \quad \left\{ \begin{array}{l} \text{sym.} \Rightarrow (b, a) \in R \\ \text{antisym.} \Rightarrow (b, a) \notin R \end{array} \right.$$

故若 R 中沒有 (a, b) with $a \neq b$ 即可同時滿足

eg. Let $A = \{1, 2, 3\}$, give a relation R on A s.t.
 R is both symmetric and antisymmetric, but
not reflexive.

Sol :

$$R = \{ (1, 1), (2, 2) \}$$

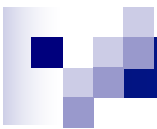


Def 5. A relation R on a set A is called **transitive(遞移)** if for $a, b, c \in A$,
 $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$.

Remark: Using quantifiers we see that the relation R on a set A is transitive if we have $\forall a \forall b \forall c ((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R$.

Example 15. Is the “divides” relation on the set of **positive integers** transitive?

Sol : Suppose $a|b$ and $b|c$
let there are positive integers k and l
such that $b=ak$ and $c=bl=(ak)l$
hence, $c=a(kl) \Rightarrow a|c$
 $\Rightarrow$ transitive



Example 13. Which of the relations in Example 7 are transitive ?

$$R_2 = \{ (1,1), (1,2), (2,1) \}$$

$$R_3 = \{ (1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4) \}$$

$$R_4 = \{ (2,1), (3,1), (3,2), (4,1), (4,2), (4,3) \}$$

Sol :

R_2 is not transitive since

$$(2,1) \in R_2 \text{ and } (1,2) \in R_2 \text{ but } (2,2) \notin R_2.$$

R_3 is not transitive since

$$(2,1) \in R_3 \text{ and } (1,4) \in R_3 \text{ but } (2,4) \notin R_3.$$

R_4 is transitive.



Example 16. How many reflexive relation are there on a set with n elements?

Sol : A relation R on a set A is a subset of $A \times A$.

$\Rightarrow A \times A$ has n^2 elements

$\Rightarrow R$ contains $(a, a) \forall a \in A$ since R is reflexive

$\Rightarrow$ There are 2^{n^2-n} reflexive relations.

Combining Relations

Example 17. Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$.

The relation $R_1 = \{(1,1), (2,2), (3,3)\}$
and $R_2 = \{(1,1), (1,2), (1,3), (1,4)\}$ can be
combined to obtain

$$R_1 \cup R_2 = \{(1,1), (2,2), (3,3), (1,2), (1,3), (1,4)\}$$

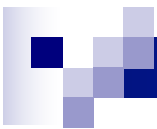
$$R_1 \cap R_2 = \{(1,1)\}$$

$$R_1 - R_2 = \{(2,2), (3,3)\}$$

$$R_2 - R_1 = \{(1,2), (1,3), (1,4)\}$$

$$R_1 \oplus R_2 = \{(2,2), (3,3), (1,2), (1,3), (1,4)\}$$

symmetric difference, 即 $(R_1 \cup R_2) - (R_1 \cap R_2)$



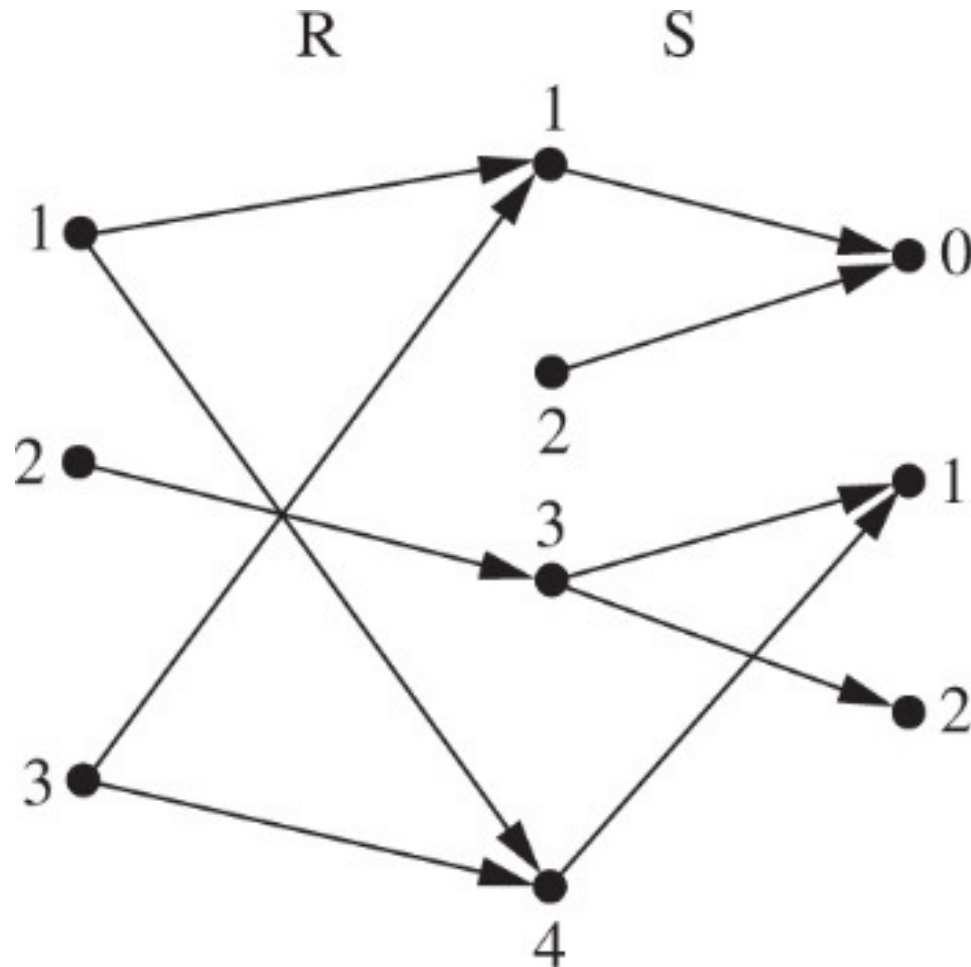
Def 6. Let R be a relation from a set A to a set B and S a relation from B to a set C . The **composite (合成) of R and S** is the relation consisting of ordered pairs (a, c) , where $a \in A$, $c \in C$, and for which there exists an element $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$. We denote the composite of R and S by $S \circ R$.

Example 20. What is the composite of relations R and S , where R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$?

Sol. $S \circ R$ is the relation from $\{1, 2, 3\}$ to $\{0, 1, 2\}$ with $S \circ R = \{(1, 0), (1, 1), (2, 1), (2, 2), (3, 0), (3, 1)\}$.

$$R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$$

$$S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$$



$$1 \rightarrow 1 \rightarrow 0 \quad (1, 0)$$

$$1 \rightarrow 4 \rightarrow 1 \quad (1, 1)$$

$$2 \rightarrow 3 \rightarrow 1 \quad (2, 1)$$

$$2 \rightarrow 3 \rightarrow 2 \quad (2, 2)$$

$$3 \rightarrow 1 \rightarrow 0 \quad (3, 0)$$

$$3 \rightarrow 4 \rightarrow 1 \quad (3, 1)$$

Sol. $S \circ R$ is the relation from $\{1, 2, 3\}$ to $\{0, 1, 2\}$ with
 $S \circ R = \{(1, 0), (1, 1), (2, 1), (2, 2), (3, 0), (3, 1)\}.$

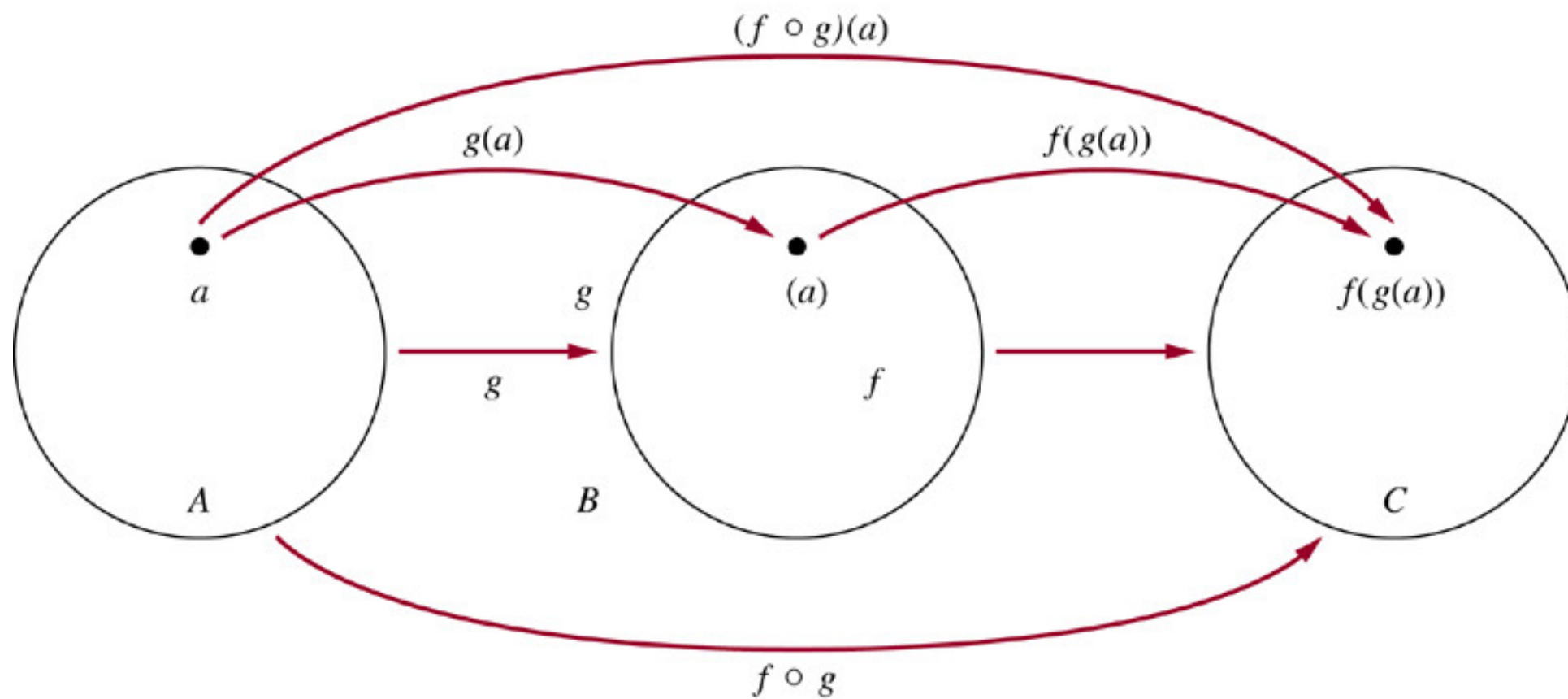


FIGURE 7 The Composition of the Functions f and g .



Def 7. Let R be a relation on the set A .

The powers R^n , $n = 1, 2, 3, \dots$, are defined recursively by $R^1 = R$ and $R^{n+1} = R^n \circ R$.

Example 22. Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$.
Find the powers R^n , $n=2, 3, 4, \dots$.

Sol. $R^2 = R \circ R = \{(1, 1), (2, 1), (3, 1), (4, 2)\}$.

$R^3 = R^2 \circ R = \{(1, 1), (2, 1), (3, 1), (4, 1)\}$.

$R^4 = R^3 \circ R = \{(1, 1), (2, 1), (3, 1), (4, 1)\} = R^3$.

Therefore $R^n = R^3$ for $n=4, 5, \dots$.

Thm 1. The relation R on a set A is transitive if and only if $R^n \subseteq R$ for $n = 1, 2, 3, \dots$.



9.2 *n*-ary Relations and their Applications

TABLE 1 Students.

<i>Student_name</i>	<i>ID_number</i>	<i>Major</i>	<i>GPA</i>
Ackermann	231455	Computer Science	3.88
Adams	888323	Physics	3.45
Chou	102147	Computer Science	3.49
Goodfriend	453876	Mathematics	3.45
Rao	678543	Mathematics	3.90
Stevens	786576	Psychology	2.99

TABLE 2 GPAs.

<i>Student_name</i>	<i>GPA</i>
Ackermann	3.88
Adams	3.45
Chou	3.49
Goodfriend	3.45
Rao	3.90
Stevens	2.99

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TABLE 3 Enrollments.

<i>Student</i>	<i>Major</i>	<i>Course</i>
Glauser	Biology	BI 290
Glauser	Biology	MS 475
Glauser	Biology	PY 410
Marcus	Mathematics	MS 511
Marcus	Mathematics	MS 603
Marcus	Mathematics	CS 322
Miller	Computer Science	MS 575
Miller	Computer Science	CS 455

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TABLE 4 Majors.

<i>Student</i>	<i>Major</i>
Glauser	Biology
Marcus	Mathematics
Miller	Computer Science

TABLE 5 Teaching_assignments.

<i>Professor</i>	<i>Department</i>	<i>Course_ number</i>
Cruz	Zoology	335
Cruz	Zoology	412
Farber	Psychology	501
Farber	Psychology	617
Grammer	Physics	544
Grammer	Physics	551
Rosen	Computer Science	518
Rosen	Mathematics	575

TABLE 6 Class_schedule.

<i>Department</i>	<i>Course_ number</i>	<i>Room</i>	<i>Time</i>
Computer Science	518	N521	2:00 P.M.
Mathematics	575	N502	3:00 P.M.
Mathematics	611	N521	4:00 P.M.
Physics	544	B505	4:00 P.M.
Psychology	501	A100	3:00 P.M.
Psychology	617	A110	11:00 A.M.
Zoology	335	A100	9:00 A.M.
Zoology	412	A100	8:00 A.M.

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TABLE 7 Teaching_schedule.

<i>Professor</i>	<i>Department</i>	<i>Course_ number</i>	<i>Room</i>	<i>Time</i>
Cruz	Zoology	335	A100	9:00 A.M.
Cruz	Zoology	412	A100	8:00 A.M.
Farber	Psychology	501	A100	3:00 P.M.
Farber	Psychology	617	A110	11:00 A.M.
Grammer	Physics	544	B505	4:00 P.M.
Rosen	Computer Science	518	N521	2:00 P.M.
Rosen	Mathematics	575	N502	3:00 P.M.

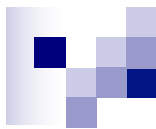


TABLE 8 (9.2)

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TABLE 8 Flights.

<i>Airline</i>	<i>Flight_number</i>	<i>Gate</i>	<i>Destination</i>	<i>Departure_time</i>
Nadir	122	34	Detroit	08:10
Acme	221	22	Denver	08:17
Acme	122	33	Anchorage	08:22
Acme	323	34	Honolulu	08:30
Nadir	199	13	Detroit	08:47
Acme	222	22	Denver	09:10
Nadir	322	34	Detroit	09:44

TABLE 9 Part_needs.

<i>Supplier</i>	<i>Part_number</i>	<i>Project</i>
23	1092	1
23	1101	3
23	9048	4
31	4975	3
31	3477	2
32	6984	4
32	9191	2
33	1001	1

2)

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TABLE 10 Parts_inventory.

<i>Part_number</i>	<i>Project</i>	<i>Quantity</i>	<i>Color_code</i>
1001	1	14	8
1092	1	2	2
1101	3	1	1
3477	2	25	2
4975	3	6	2
6984	4	10	1
9048	4	12	2
9191	2	80	4



9.3 Representing Relations

Representing Relations using Matrices

Suppose that R is a relation from $A = \{a_1, a_2, \dots, a_m\}$ to $B = \{b_1, b_2, \dots, b_n\}$.

The relation R can be represented by the matrix

$M_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1, & \text{if } (a_i, b_j) \in R \\ 0, & \text{if } (a_i, b_j) \notin R \end{cases}$$

Example 1. Suppose that $A = \{1, 2, 3\}$ and $B = \{1, 2\}$
Let $R = \{(a, b) \mid a > b, a \in A, b \in B\}$.
What is the matrix M_R representing R ?

Sol :

$$R = \{(2, 1), (3, 1), (3, 2)\}$$

$$\begin{array}{c} A \end{array} \begin{array}{c} 1 \\ 2 \\ 3 \end{array} \begin{array}{cc} & B \\ & \begin{array}{cc} 1 & 2 \end{array} \\ \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{array} \right] \end{array} \quad \therefore M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Note. 不同的A,B的元素順序會製造不同的 M_R 。
若A=B, 行列應使用相同的順序。

※ Let $A = \{a_1, a_2, \dots, a_n\}$.

A relation R on A is reflexive iff $(a_i, a_i) \in R, \forall i$.

i.e.,

$$M_R = \begin{matrix} & \begin{matrix} a_1 & a_2 & \dots & \dots & a_n \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{matrix} & \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix} \end{matrix} \quad \text{對角線上全為1}$$

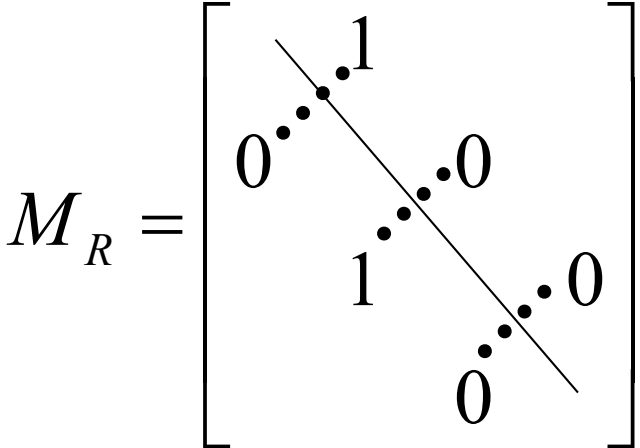
※ The relation R is symmetric iff $(a_i, a_j) \in R \Rightarrow (a_j, a_i) \in R$.

This means $m_{ij} = m_{ji}$ (即 M_R 是對稱矩陣).

$$M_R = \begin{bmatrix} & & & & 1 \\ & & & & \cdot \\ & & & & \cdot \\ & & & & \cdot \\ & & & & \cdot \\ 1 & \cdot & \cdot & \cdot & \cdot \\ & & & & 0 \\ & & & & \cdot \\ & & & & \cdot \\ & & & & \cdot \\ & & 0 & \cdot & \cdot & \cdot & \cdot \\ & & & & & & 0 \end{bmatrix} = (M_R)^t$$

※ The relation R is antisymmetric iff
 $(a_i, a_j) \in R$ and $i \neq j \Rightarrow (a_j, a_i) \notin R$.

This means that if $m_{ij}=1$ with $i \neq j$, then $m_{ji}=0$.
i.e.,

$$M_R = \begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix}$$


※ transitive 性質不易從矩陣直接判斷出來，需做運算



Example 3. Suppose that the relation R on a set is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Is R reflexive, symmetric, and/or antisymmetric ?

Sol :

reflexive, symmetric, not antisymmetric.



eg. Suppose that $S = \{0, 1, 2, 3\}$. Let R be a relation containing (a, b) if $a \leq b$, where $a \in S$ and $b \in S$. Is R reflexive, symmetric, antisymmetric?

Sol :

$$M_R = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$\therefore R$ is reflexive and antisymmetric, not symmetric.

Def. irreflexive(非反身性) : $(a, a) \notin R, \forall a \in A$



The Boolean operations join and meet (discussed in Section 3.8) can be used to find the matrices representing the union and the intersection of two relations. Suppose that R_1 and R_2 are relations on a set A represented by the matrices $\mathbf{M}_{R_1}$ and $\mathbf{M}_{R_2}$, respectively. The matrix representing the union of these relations has a 1 in the positions where either $\mathbf{M}_{R_1}$ or $\mathbf{M}_{R_2}$ has a 1. The matrix representing the intersection of these relations has a 1 in the positions where both $\mathbf{M}_{R_1}$ and $\mathbf{M}_{R_2}$ have a 1. Thus, the matrices representing the union and intersection of these relations are

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} \quad \text{and} \quad \mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2}.$$



EXAMPLE 4 Suppose that the relations R_1 and R_2 on a set A are represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$?

Solution: The matrices of these relations are

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$

$$\mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

We now turn our attention to determining the matrix for the composite of relations. This matrix can be found using the Boolean product of the matrices (discussed in Section 3.8) for these relations. In particular, suppose that R is a relation from A to B and S is a relation from B to C . Suppose that A , B , and C have m , n , and p elements, respectively. Let the zero-one matrices for $S \circ R$, R , and S be $\mathbf{M}_{S \circ R} = [t_{ij}]$, $\mathbf{M}_R = [r_{ij}]$, and $\mathbf{M}_S = [s_{ij}]$, respectively (these matrices have sizes $m \times p$, $m \times n$, and $n \times p$, respectively). The ordered pair (a_i, c_j) belongs to $S \circ R$ if and only if there is an element b_k such that (a_i, b_k) belongs to R and (b_k, c_j) belongs to S . It follows that $t_{ij} = 1$ if and only if $r_{ik} = s_{kj} = 1$ for some k . From the definition of the Boolean product, this means that

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S.$$

EXAMPLE 5 Find the matrix representing the relations $S \circ R$, where the matrices representing R and S are

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Solution: The matrix for $S \circ R$ is

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The matrix representing the composite of two relations can be used to find the matrix for $\mathbf{M}_{R^n}$. In particular,

$$\mathbf{M}_{R^n} = \mathbf{M}_R^{[n]},$$

from the definition of Boolean powers. Exercise 35 at the end of this section asks for a proof of this formula.

$M_R \times M_S$ (矩陣乘法) 之後將 >1 的數字改為1

DEFINITION 4

Let $\mathbf{A}$ be an $m \times k$ matrix and $\mathbf{B}$ be a $k \times n$ matrix. The *product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{AB}$, is the $m \times n$ matrix with its (i, j) th entry equal to the sum of the products of the corresponding elements from the i th row of $\mathbf{A}$ and the j th col

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj}.$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ik} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mk} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2n} \\ \vdots & & & \vdots & & \vdots \\ b_{k1} & b_{k2} & \cdots & b_{kj} & \cdots & b_{kn} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & & c_{ij} & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{bmatrix}$$

EXAMPLE 6

Find the matrix representing the relation R^2 , where the matrix representing R is

$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

Solution: The matrix for R^2 is

$$\mathbf{M}_{R^2} = \mathbf{M}_R^{[2]} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$



EXAMPLE 7 The directed graph with vertices a , b , c , and d , and edges (a, b) , (a, d) , (b, b) , (b, d) , (c, a) , (c, b) , and (d, b) is displayed in Figure 3.

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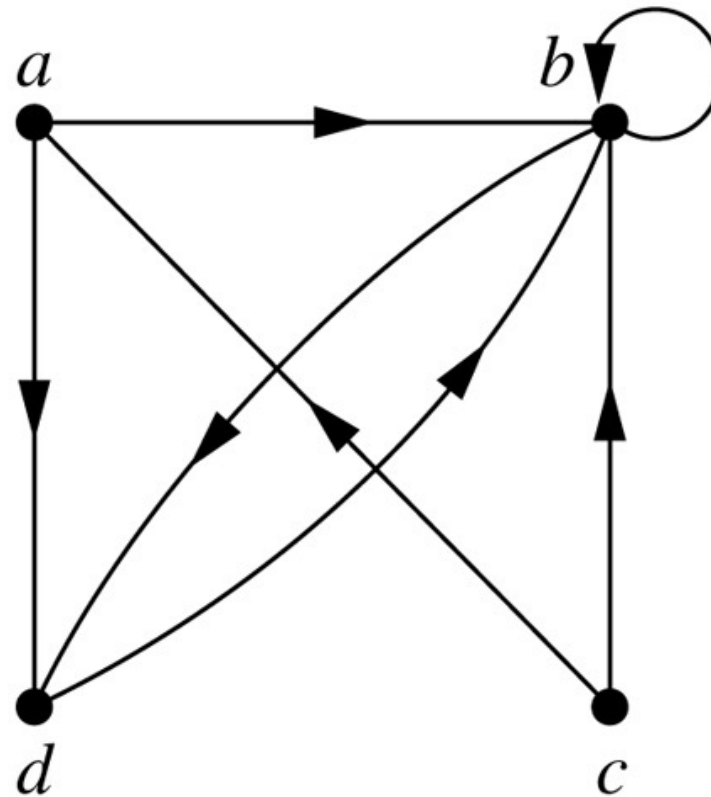


FIGURE 3 The Directed Graph.

※Representing Relations using Digraphs. (directed graphs)

Example 8. Show the directed graph (digraph) of the relation $R = \{(1,1), (1,3), (2,1), (2,3), (2,4), (3,1), (3,2), (4,1)\}$ on the set $\{1,2,3,4\}$.

Sol :

vertex(點) : 1, 2, 3, 4

edge(邊) : $1 \rightarrow 1, 1 \rightarrow 3, 2 \rightarrow 1$
 $2 \rightarrow 3, 2 \rightarrow 4, 3 \rightarrow 1$
 $3 \rightarrow 2, 4 \rightarrow 1$

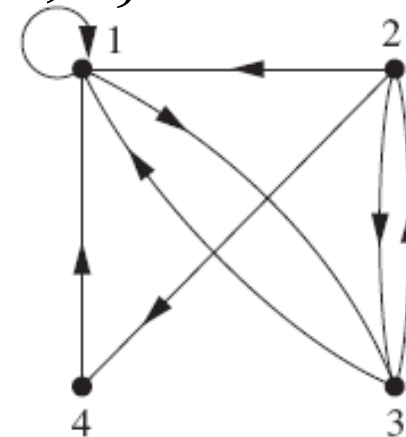


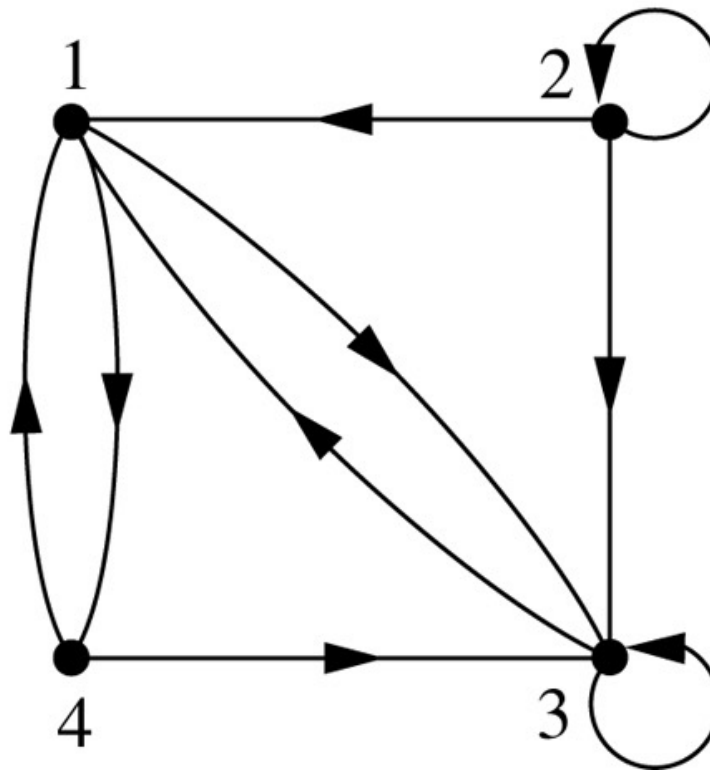
FIGURE 4 The Directed Graph of the Relation R .

EXAMPLE 9 What are the ordered pairs in the relation R represented by the directed graph shown in Figure 5?

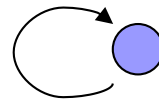
Solution: The ordered pairs (x, y) in the relation are

$$R = \{(1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (3, 1), (3, 3), (4, 1), (4, 3)\}.$$

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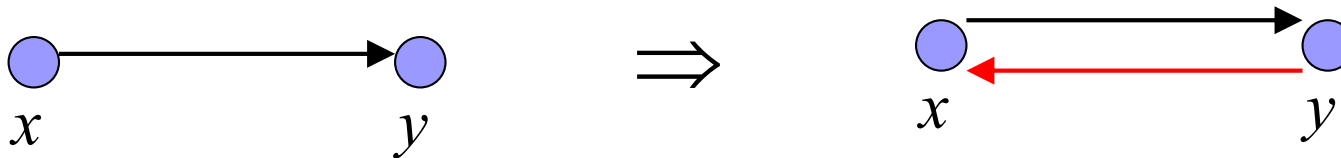


※ The relation R is reflexive iff
for every vertex,



(每個點上都有loop)

※ The relation R is symmetric iff



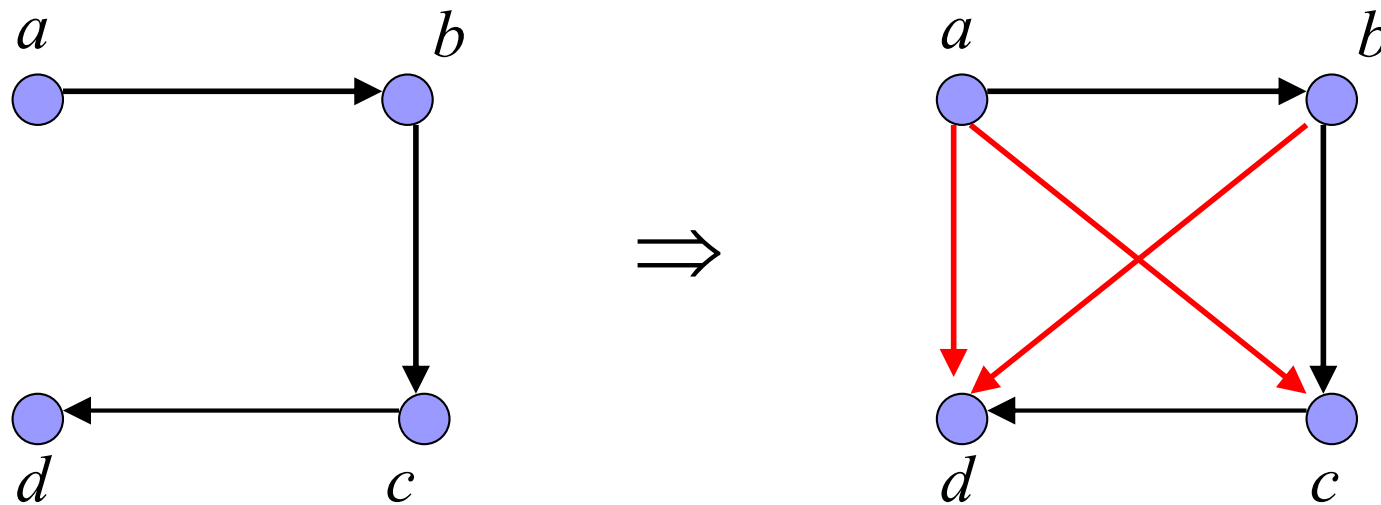
(兩點間若有邊，必為一對不同方向的邊)

※ The relation R is antisymmetric iff

兩點間若有邊，必只有一條邊

※ The relation R is transitive iff
for $a, b, c \in A$,
 $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$.

This means:



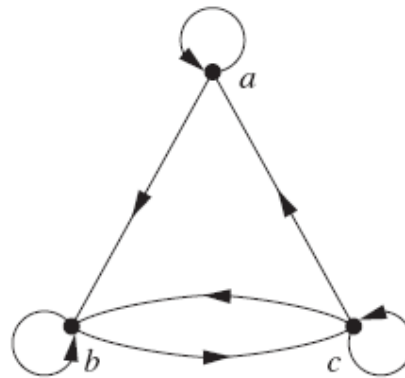
(只要點 x 有路徑走到點 y ， x 必定有邊直接連向 y)

Example 10. Determine whether the relations R and S are reflexive, symmetric, antisymmetric, and/or transitive

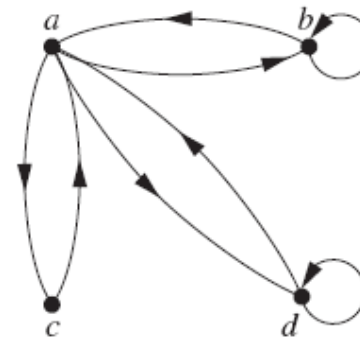
Sol :

reflexive,
not symmetric,
not antisymmetric,
not transitive

$(a \rightarrow b, b \rightarrow c, a \not\rightarrow c)$



(a) Directed graph of R



(b) Directed graph of S

not reflexive,
symmetric
not antisymmetric
not transitive

$(b \rightarrow a, a \rightarrow c, b \not\rightarrow c)$

FIGURE 6 The Directed Graphs of the Relations R and S .

irreflexive(非反身性)的定義在 p.557
即 $(a,a) \notin R, \forall a \in A$



14. Let R_1 and R_2 be relations on a set A represented by the matrices

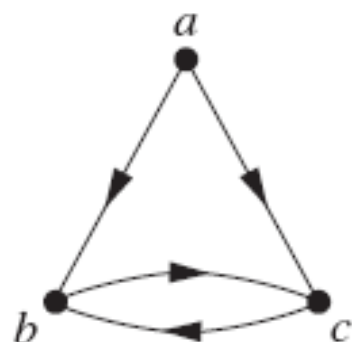
$$\mathbf{M}_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Find the matrices that represent

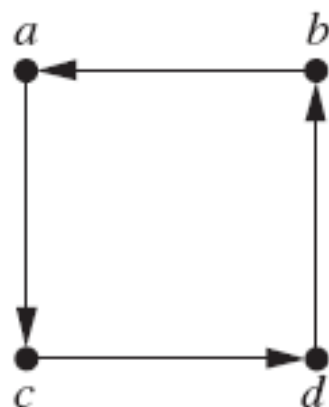
- a)** $R_1 \cup R_2$. **b)** $R_1 \cap R_2$. **c)** $R_2 \circ R_1$.
d) $R_1 \circ R_1$. **e)** $R_1 \oplus R_2$.

In Exercises 23–28 list the ordered pairs in the relations represented by the directed graphs.

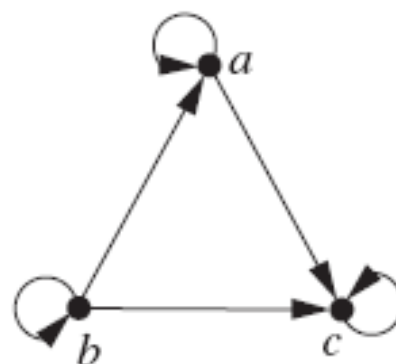
23.



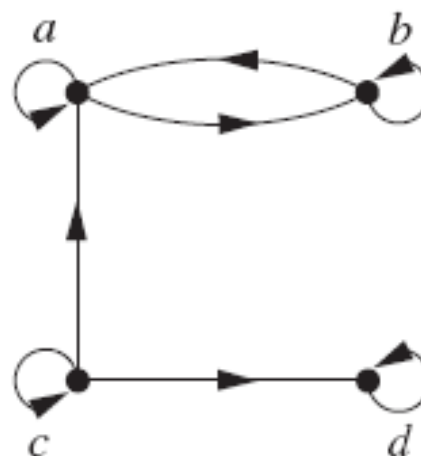
25.



24.



26.





9.4 Closures of Relations

✂ Closures

The relation $R = \{(1,1), (1,2), (2,1), (3,2)\}$ on the set $A = \{1, 2, 3\}$ is not reflexive.

Q: How to construct a smallest reflexive relation R_r such that $R \subseteq R_r$?

Sol: Let $R_r = R \cup \{(2,2), (3,3)\}$.

i. e., $R_r = R \cup \{(a, a) \mid a \in A\}$.

R_r is a reflexive relation containing R that is as small as possible. It is called the **reflexive closure** of R .



Example 1. What is the reflexive closure of the relation $R = \{(a, b) \mid a < b\}$ on the set of integers ?

Sol :

$$\begin{aligned} R_r &= R \cup \{ (a, a) \mid a \in \mathbf{Z} \} \\ &= \{ (a, b) \mid a \leq b, \ a, b \in \mathbf{Z} \} \end{aligned}$$

Example :

The relation $R = \{ (1, 1), (1, 2), (2, 2), (2, 3), (3, 1), (3, 2) \}$ on the set $A = \{1, 2, 3\}$ is not symmetric.

Sol :

Let $R^{-1} = \{ (a, b) \mid (a, b) \in R \}$

Let $R_s = R \cup R^{-1} = \{ (1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 1), (1, 3), (3, 2) \}$

R_s is the smallest symmetric relation containing R , called the symmetric closure of R .



Example 2. What is the symmetric closure of the relation $R = \{(a, b) \mid a > b\}$ on the set of positive integers ?

Sol :

$$R \cup \{(b, a) \mid a > b\} = \{(a, b) \mid a \neq b\}$$

$$\parallel$$

$$\{(a, b) \mid a < b\}$$

Def :

1. (reflexive closure of R on A)

R_r = the smallest set containing R and is reflexive.

$$R_r = R \cup \{ (a, a) \mid a \in A, (a, a) \notin R \}$$

2. (symmetric closure of R on A)

R_s = the smallest set containing R and is symmetric

$$R_s = R \cup \{ (b, a) \mid (a, b) \in R \text{ \& } (b, a) \notin R \}$$

3. (transitive closure of R on A)

R_t = the smallest set containing R and is transitive.

$$R_t = R \cup \{ (a, c) \mid (a, b) \in R \text{ \& } (b, c) \in R, \text{ but } (a, c) \notin R \}$$

Note. 沒有antisymmetric closure，因若不是antisymmetric，表示有 $a \neq b$ ，且 (a, b) 及 (b, a) 都 $\in R$ ，此時加任何pair都不可能變成 antisymmetric.

EXAMPLE 3 Which of the following are paths in the directed graph shown in Figure 1: a, b, e, d ; a, e, c, d, b ; b, a, c, b, a, a, b ; d, c ; c, b, a ; e, b, a, b, a, b, e ? What are the lengths of those that are paths? Which of the paths in this list are circuits?

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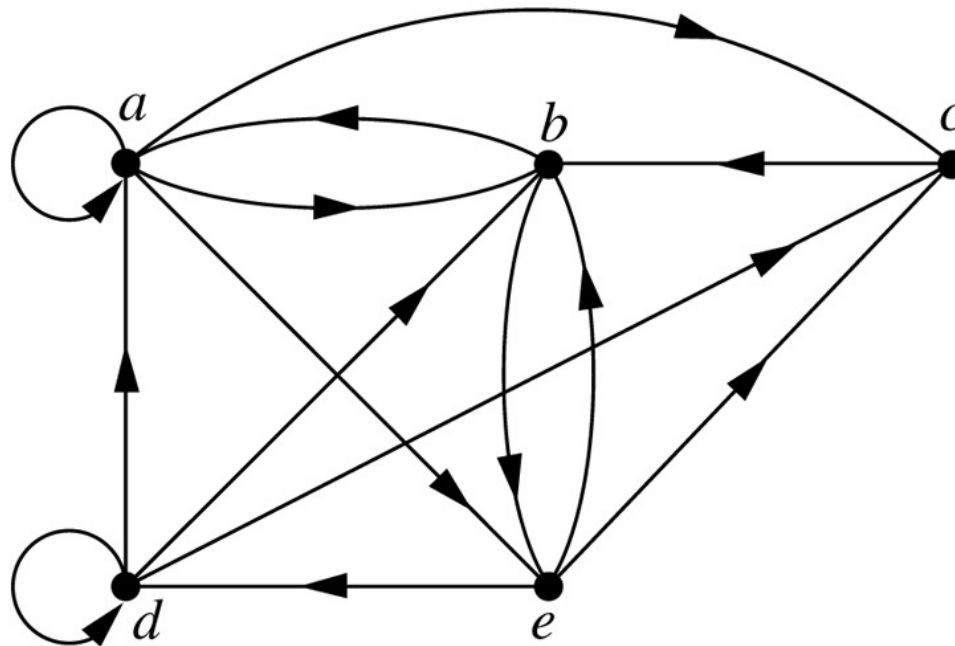
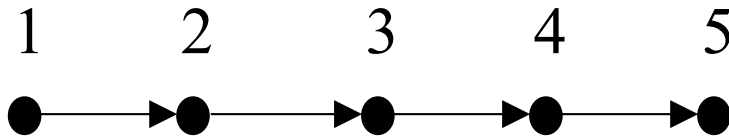


FIGURE 1 A Directed Graph.

Paths in Directed Graphs

Def 1. A **path** from a to b in the digraph G is a sequence of edges $(x_0, x_1), (x_1, x_2), \dots, (x_{n-1}, x_n)$ in G , where $n \in \mathbb{Z}^+$, and $x_0 = a, x_n = b$. This path is denoted by $x_0, x_1, x_2, \dots, x_n$ and has **length** n .

Ex.



A **path** from 1 to 5
of length 4

Theorem 1 Let R be a relation on a set A . There is a path of length n , where $n \in \mathbb{Z}^+$, from a to b if and only if $(a, b) \in R^n$.



Transitive Closures

We now show that finding the transitive closure of a relation is equivalent to determining which pairs of vertices in the associated directed graph are connected by a path. With this in mind, we define a new relation.

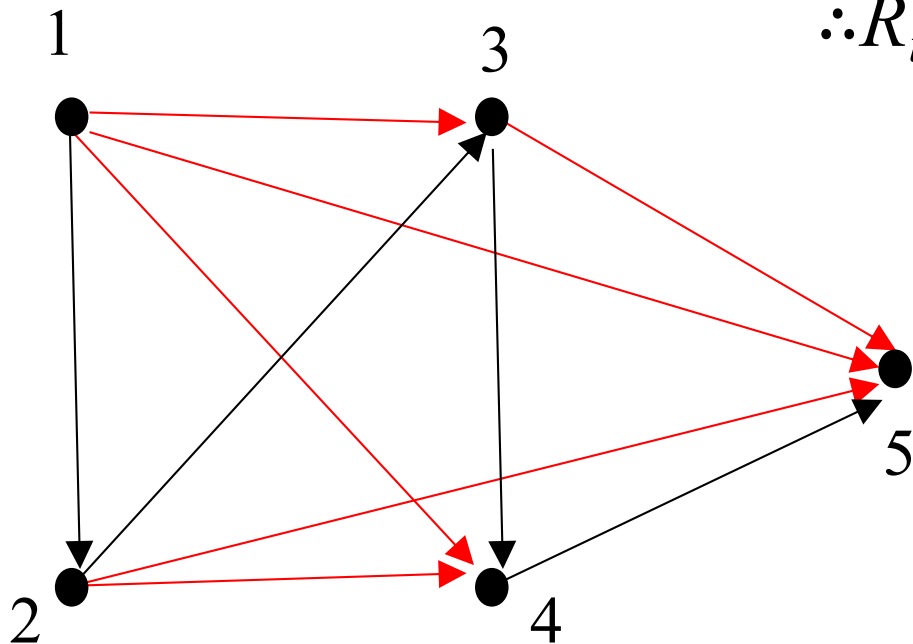
DEFINITION 2 Let R be a relation on a set A . The *connectivity relation* R^* consists of the pairs (a, b) such that there is a path of length at least one from a to b in R .

Because R^n consists of the pairs (a, b) such that there is a path of length n from a to b , it follows that R^* is the union of all the sets R^n . In other words,

$$R^* = \bigcup_{n=1}^{\infty} R^n.$$

Example. Let R be a relation on a set A , where
 $A = \{1, 2, 3, 4, 5\}$, $R = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$.
What is the transitive closure R_t of R ?

Sol :

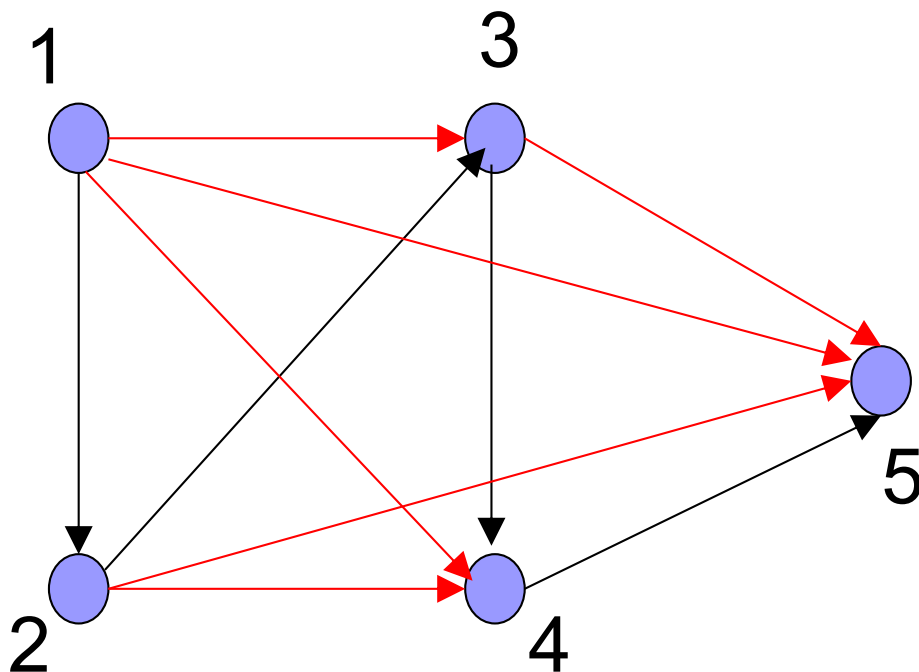


$$\begin{aligned}\therefore R_t &= R \cup R^2 \cup R^3 \cup R^4 \cup R^5 \\ &= \{(1, 2), (2, 3), (3, 4), (4, 5), \\ &\quad (1, 3), (2, 4), (3, 5), \\ &\quad (1, 4), (2, 5), \\ &\quad (1, 5)\}\end{aligned}$$

Example. Let R be a relation on a set A , where
 $A = \{1, 2, 3, 4, 5\}$, $R = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$.
What is the transitive closure R_t of R ?

Sol :

$$\therefore R_t = \left\{ \begin{array}{l} (1, 2), (2, 3), (3, 4), (4, 5) \\ (1, 3), (1, 4), (1, 5) \\ (2, 4), (2, 5) \\ (3, 5) \end{array} \right\}$$



另法： $M_{R^t} = M_R \vee M_R^{[2]} \vee \dots \vee M_R^{[n]}$,
where $n = |A|$.

其中，[2]表示2次方， $\vee$ 運算類似加法，但兩者都會把 ≥ 1 改為1。



Theorem 3 Let M_R be the zero-one matrix of the relation R on a set with n elements. Then the zero-one matrix of the transitive closure R^* is

$$M_{R^*} = M_R \vee M_R^{[2]} \vee \cdots \vee M_R^{[n]}.$$

Example 7. Find the zero-one matrix of the transitive closure of the relation R where

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

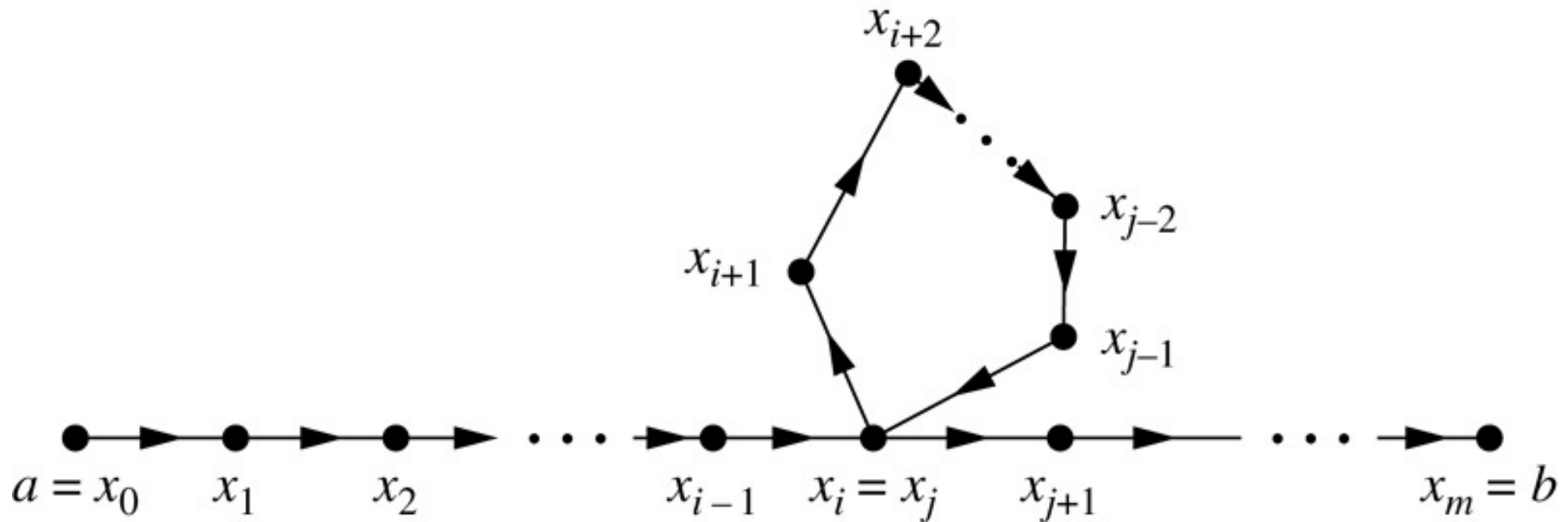
Sol :

$$M_{R^*} = M_R \vee M_R^{[2]} \vee M_R^{[3]} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

LEMMA 1

Let A be a set with n elements, and let R be a relation on A . If there is a path of length at least one in R from a to b , then there is such a path with length not exceeding n . Moreover, when $a \neq b$, if there is a path of length at least one in R from a to b , then there is such a path with length not exceeding $n - 1$.

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From Lemma 1, we see that the transitive closure of R is the union of $R, R^2, R^3, \dots$, and R^n . This follows because there is a path in R^* between two vertices if and only if there is a path between these vertices in R^i , for some positive integer i with $i \leq n$. Because

$$R^* = R \cup R^2 \cup R^3 \cup \dots \cup R^n$$

EXAMPLE 7 Find the zero-one matrix of the transitive closure of the relation R where

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution: From Theorem 3, it follows that the zero-one matrix of R^* is

$$\mathbf{M}_{R^*} = \mathbf{M}_R \vee \mathbf{M}_R^{[2]} \vee \mathbf{M}_R^{[3]}.$$

Because

$$\mathbf{M}_R^{[2]} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_R^{[3]} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix},$$

it follows that

$$\mathbf{M}_{R^*} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

ALGORITHM 1 A Procedure for Computing the Transitive Closure.

procedure *transitive closure* ($\mathbf{M}_R$: zero-one $n \times n$ matrix)

$\mathbf{A} := \mathbf{M}_R$

$\mathbf{B} := \mathbf{A}$

for $i := 2$ **to** n

begin

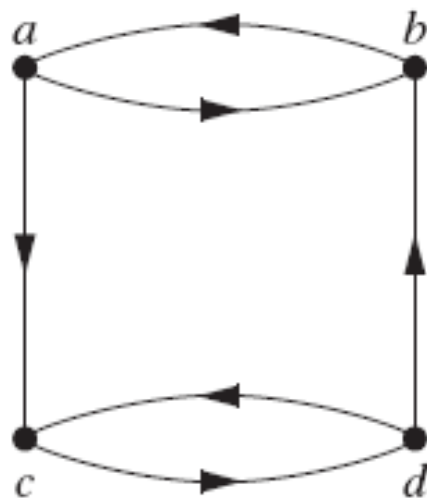
$\mathbf{A} := \mathbf{A} \odot \mathbf{M}_R$

$\mathbf{B} := \mathbf{B} \vee \mathbf{A}$

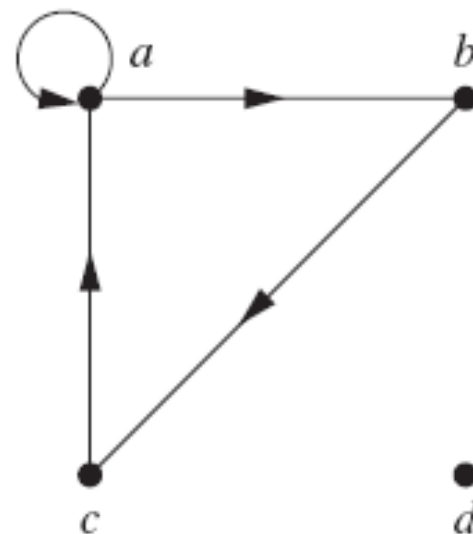
end { $\mathbf{B}$ is the zero-one matrix for R^* }

In Exercises 5–7 draw the directed graph of the reflexive closure of the relations with the directed graph shown.

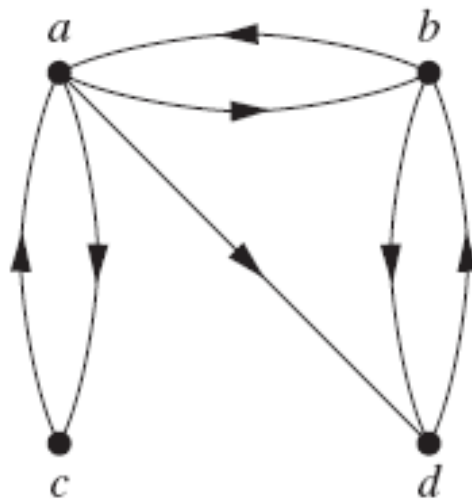
5.



6.

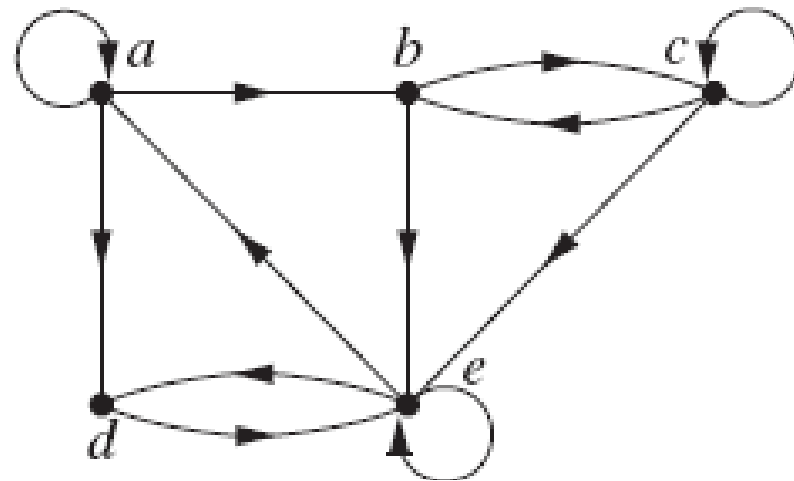


7.



16. Determine whether these sequences of vertices are paths in this directed graph.

- a) a, b, c, e
- b) b, e, c, b, e
- c) a, a, b, e, d, e
- d) b, c, e, d, a, a, b
- e) b, c, c, b, e, d, e, d
- f) $a, a, b, b, c, c, b, e, d$



17. Find all circuits of length three in the directed graph in Exercise 16.

18. Determine whether there is a path in the directed graph in Exercise 16 beginning at the first vertex given and ending at the second vertex given.

- | | | |
|-----------|-----------|-----------|
| a) a, b | b) b, a | c) b, b |
| d) a, e | e) b, d | f) c, d |
| g) d, d | h) e, a | i) e, c |



9.5 Equivalence Relations (等價關係)

Def 1. A relation R on a set A is called an **equivalence relation** if it is reflexive, symmetric, and transitive.

Example 1.

Let R be the relation on the set of integers such that aRb if and only if $a=b$ or $a=-b$. Then R is an equivalence relation.

Example 2.

Let R be the relation on the set of real numbers such that aRb if and only if $a-b$ is an integer. Then R is an equivalence relation.

Example 3. (Congruence Modulo m)

Let $m \in \mathbf{Z}$ and $m > 1$. Show that the relation $R = \{ (a, b) \mid a \equiv b \pmod{m} \}$ is an equivalence relation on the set of integers.

(a is congruent to b modulo m)

Sol : Note that $a \equiv b \pmod{m}$ iff $m \mid (a - b)$.

$\therefore$ ① $a \equiv a \pmod{m} \Rightarrow (a, a) \in R \Rightarrow$ **reflexive**

② If $a \equiv b \pmod{m}$, then $a - b = km, k \in \mathbf{Z}$

$\Rightarrow b - a = (-k)m \Rightarrow b \equiv a \pmod{m} \Rightarrow$ **symmetric**

③ If $a \equiv b \pmod{m}$, $b \equiv c \pmod{m}$

then $a - b = km, b - c = lm$

$\Rightarrow a - c = (k + l)m \Rightarrow a \equiv c \pmod{m} \Rightarrow$ **transitive**

$\therefore R$ is an equivalence relation.

9.5 Equivalence Relations (等價關係)

Def 1. A relation R on a set A is called an **equivalence relation** if it is reflexive, symmetric, and transitive.

Example 4.

Let $l(x)$ denote the length of the string x .

Suppose that the relation

$R = \{(a, b) \mid l(a) = l(b), a, b \text{ are strings of English letters} \}$

Is R an equivalence relation ?

Sol :

- | | | |
|---|--------------------------|--------|
| ① $(a, a) \in R \quad \forall \text{string } a$ | $\Rightarrow$ reflexive | } Yes. |
| ② $(a, b) \in R \Rightarrow (b, a) \in R$ | $\Rightarrow$ symmetric | |
| ③ $(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$ | $\Rightarrow$ transitive | |



EXAMPLE 5 Let n be a positive integer and S a set of strings. Suppose that R_n is the relation on S such that sR_nt if and only if $s = t$, or both s and t have at least n characters and the first n characters of s and t are the same. That is, a string of fewer than n characters is related only to itself; a string s with at least n characters is related to a string t if and only if t has at least n characters and t begins with the n characters at the start of s . For example, let $n = 3$ and let S be the set of all bit strings. Then sR_3t either when $s = t$ or both s and t are bit strings of length 3 or more that begin with the same three bits. For instance, $01R_301$ and $00111R_300101$, but $01 \not R_3 010$ and $01011 \not R_3 01110$.

Show that for every set S of strings and every positive integer n , R_n is an equivalence relation on S .



EXAMPLE 6 Show that the “divides” relation on the set of positive integers is not an equivalence relation.

Solution: By Examples 9 and 15 in Section 8.1, we know that the “divides” relation is reflexive and transitive. However, by Example 12 in Section 8.1, we know that this relation is not symmetric (for instance, $2 \mid 4$ but $4 \nmid 2$). We conclude that the “divides” relation on the set of positive integers is not an equivalence relation. 

EXAMPLE 7 Let R be the relation on the set of real numbers such that $x R y$ if and only if x and y are real numbers that differ by less than 1, that is $|x - y| < 1$. Show that R is not an equivalence relation.

Solution: R is reflexive because $|x - x| = 0 < 1$ whenever $x \in \mathbf{R}$. R is symmetric, for if $x R y$, where x and y are real numbers, then $|x - y| < 1$, which tells us that $|y - x| = |x - y| < 1$, so that $y R x$. However, R is not an equivalence relation because it is not transitive. Take $x = 2.8$, $y = 1.9$, and $z = 1.1$, so that $|x - y| = |2.8 - 1.9| = 0.9 < 1$, $|y - z| = |1.9 - 1.1| = 0.8 < 1$, but $|x - z| = |2.8 - 1.1| = 1.7 > 1$. That is, $2.8 R 1.9$, $1.9 R 1.1$, but $2.8 \not R 1.1$. 

Example 7.

Let R be the relation on the set of real numbers such that xRy if and only if x and y differ by less than 1, that is $|x - y| < 1$. Show that R is not an equivalence relation.

Sol :

$$\textcircled{1} \ xRx \ \forall x \text{ since } x - x = 0 \quad \Rightarrow \text{reflexive}$$

$$\textcircled{2} \ xRy \Rightarrow |x - y| < 1 \Rightarrow |y - x| < 1 \Rightarrow yRx \\ \Rightarrow \text{symmetric}$$

$$\textcircled{3} \ xRy, yRz \Rightarrow |x - y| < 1, |y - z| < 1 \quad \not\Rightarrow |x - z| < 1 \\ \Rightarrow \text{Not transitive}$$



Equivalence Classes

Def 3.

Let R be an equivalence relation on a set A .

The equivalence class of the element $a \in A$ is

$$[a]_R = \{ s \mid (a, s) \in R \}$$

For any $b \in [a]_R$, b is called a representative of this equivalence class.

Note:

If $(a, b) \in R$, then $[a]_R = [b]_R$.



Example 9.

What are the equivalence class of 0 and 1 for congruence modulo 4 ?

Sol :

Let $R = \{ (a,b) \mid a \equiv b \pmod{4} \}$

Then $[0]_R = \{ s \mid (0,s) \in R \}$
 $= \{ \dots, -8, -4, 0, 4, 8, \dots \}$

$[1]_R = \{ t \mid (1,t) \in R \} = \{ \dots, -7, -3, 1, 5, 9, \dots \}$

Congruence classes modulo m :

$$[a]_m = \{ \dots, a-2m, a-m, a, a+m, a+2m, \dots \}$$



EXAMPLE 10 What is the equivalence class of the string 0111 with respect to the equivalence relation R_3 from Example 5 on the set of all bit strings? (Recall that sR_3t if and only if s and t are bit strings with $s = t$ or s and t are strings of at least three bits that start with the same three bits.)

Solution: The bit strings equivalent to 0111 are the bit strings with at least three bits that begin with 011. These are the bit strings 011, 0110, 0111, 01100, 01101, 01110, 01111, and so on. Consequently,

$$[011]_{R_3} = \{011, 0110, 0111, 01100, 01101, 01110, 01111, \dots\}.$$




THEOREM 1

Let R be an equivalence relation on a set A . These statements for elements a and b of A are equivalent:

$$(i) aRb \quad (ii) [a] = [b] \quad (iii) [a] \cap [b] \neq \emptyset$$

Def.

A partition (分割) of a set S is a collection of disjoint nonempty subsets A_i of S that have S as their union.

In other words, we have

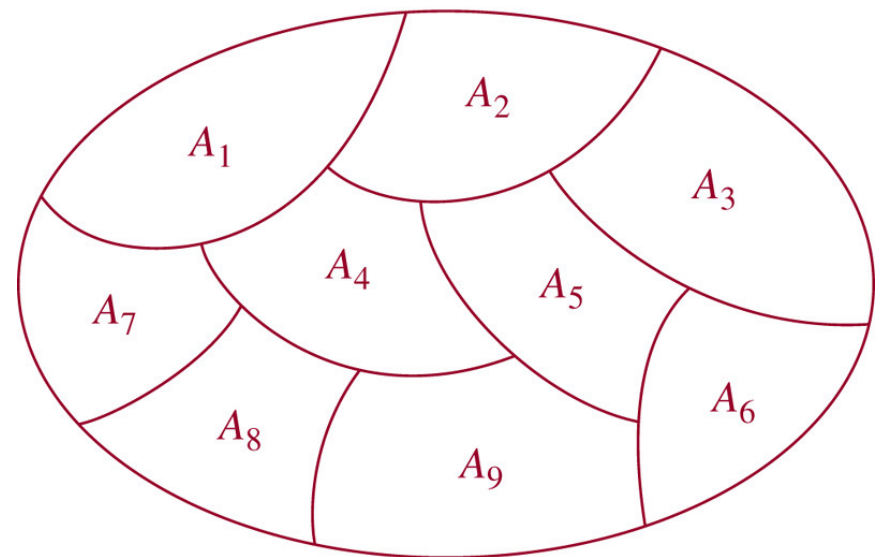
$$A_i \neq \emptyset, \quad \forall i,$$

$$A_i \cap A_j = \emptyset, \quad \text{when } i \neq j$$

and

$$\cup A_i = S.$$

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Equivalence Classes and Partitions

Def.

A partition (分割) of a set S is a collection of disjoint nonempty subsets A_i of S that have S as their union.

In other words, we have $A_i \neq \emptyset, \forall i$,

$$A_i \cap A_j = \emptyset, \text{ for any } i \neq j, \text{ and } \cup A_i = S.$$

Example 12.

Suppose that $S = \{1, 2, 3, 4, 5, 6\}$. The collection of sets $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 5\}$, and $A_3 = \{6\}$ form a partition of S .



Thm 2.

Let R be an equivalence relation on a set A .

Then the equivalence classes of R form a partition of A .

Example 13.

List the ordered pairs in the equivalence relation R produced by the partition $A_1=\{1, 2, 3\}$, $A_2=\{4, 5\}$, and $A_3=\{6\}$ of $S=\{1, 2, 3, 4, 5, 6\}$.

Sol :

$$\begin{aligned} R &= \{ (a, b) \mid a, b \in A_1 \} \cup \{ (a, b) \mid a, b \in A_2 \} \\ &\quad \cup \{ (a, b) \mid a, b \in A_3 \} \\ &= \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), \\ &\quad (3, 3), (4, 4), (4, 5), (5, 4), (5, 5), (6, 6)\} \end{aligned}$$

Example 14. $[a]_m = \{..., a - 2m, a - m, a, a + m, a + 2m, ...\}$

The congruence modulo 4 form a partition of the integers.

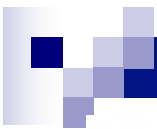
Sol :

$$[0]_4 = \{ ..., -8, -4, 0, 4, 8, ... \}$$

$$[1]_4 = \{ ..., -7, -3, 1, 5, 9, ... \}$$

$$[2]_4 = \{ ..., -6, -2, 2, 6, 10, ... \}$$

$$[3]_4 = \{ ..., -5, -1, 3, 7, 11, ... \}$$



EXAMPLE 15 Let R_3 be the relation from Example 5. What are the sets in the partition of the set of all bit strings arising from the relation R_3 on the set of all bit strings? (Recall that $s R_3 t$, where s and t are bit strings, if $s = t$ or s and t are bit strings with at least three bits that agree in their first three bits.)

Solution: Note that every bit string of length less than three is equivalent only to itself. Hence $[\lambda]_{R_3} = \{\lambda\}$, $[0]_{R_3} = \{0\}$, $[1]_{R_3} = \{1\}$, $[00]_{R_3} = \{00\}$, $[01]_{R_3} = \{01\}$, $[10]_{R_3} = \{10\}$, and $[11]_{R_3} = \{11\}$. Note that every bit string of length three or more is equivalent to one of the eight bit strings 000, 001, 010, 011, 100, 101, 110, and 111. We have

$$[000]_{R_3} = \{000, 0000, 0001, 00000, 00001, 00010, 00011, \dots\},$$

$$[001]_{R_3} = \{001, 0010, 0011, 00100, 00101, 00110, 00111, \dots\},$$

$$[010]_{R_3} = \{010, 0100, 0101, 01000, 01001, 01010, 01011, \dots\},$$

$$[011]_{R_3} = \{011, 0110, 0111, 01100, 01101, 01110, 01111, \dots\},$$

$$[100]_{R_3} = \{100, 1000, 1001, 10000, 10001, 10010, 10011, \dots\},$$

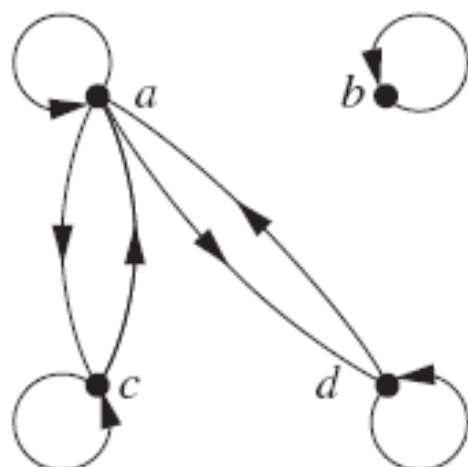
$$[101]_{R_3} = \{101, 1010, 1011, 10100, 10101, 10110, 10111, \dots\},$$

$$[110]_{R_3} = \{110, 1100, 1101, 11000, 11001, 11010, 11011, \dots\}, \text{ and}$$

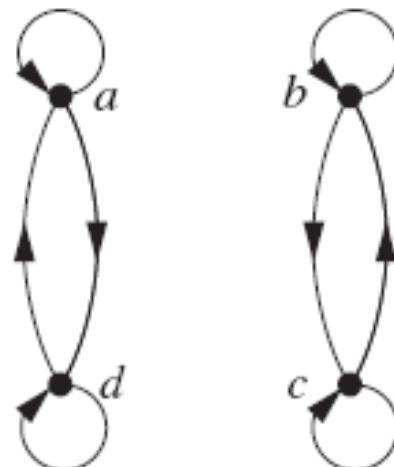
$$[111]_{R_3} = \{111, 1110, 1111, 11100, 11101, 11110, 11111, \dots\}.$$

In Exercises 21–23 determine whether the relation with the directed graph shown is an equivalence relation.

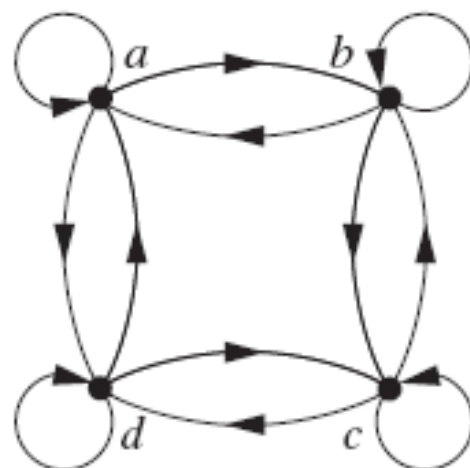
21.



22.



23.





24. Determine whether the relations represented by these zero-one matrices are equivalence relations.

a) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ **b)** $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$ **c)** $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

41. Which of these collections of subsets are partitions of $\{1, 2, 3, 4, 5, 6\}$?

- a)** $\{1, 2\}, \{2, 3, 4\}, \{4, 5, 6\}$ **b)** $\{1\}, \{2, 3, 6\}, \{4\}, \{5\}$
c) $\{2, 4, 6\}, \{1, 3, 5\}$ **d)** $\{1, 4, 5\}, \{2, 6\}$

9.6 Partial Orderings

Def 1. A relation R on a set S is called a **partial ordering** (偏序) or **partial order** if it is reflexive, **antisymmetric**, and transitive. A set S together with a partial ordering R is called a **partially ordered set**, or **poset**, and is denoted by (S, R) .

Example 1.

Show that the “greater than or equal” ($\geq$) is a partial ordering on the set of integers.

Sol :

- ① $x \geq x \quad \forall x \in \mathbf{Z}$ $\Rightarrow$ reflexive
- ② If $x \geq y$ and $y \geq x$ then $x = y$. $\Rightarrow$ antisymmetric
- ③ $x \geq y, y \geq z \Rightarrow x \geq z$ $\Rightarrow$ transitive

9.6 Partial Orderings (偏序)

DEFINITION 1

A relation R on a set S is called a *partial ordering* or *partial order* if it is reflexive, antisymmetric, and transitive. A set S together with a partial ordering R is called a *partially ordered set*, or *poset*, and is denoted by (S, R) . Members of S are called *elements* of the poset.

EXAMPLE 2

The divisibility relation $|$ is a partial ordering on the set of positive integers, because it is reflexive, antisymmetric, and transitive, as was shown in Section 8.1. We see that $(\mathbf{Z}^+, |)$ is a poset. Recall that $(\mathbf{Z}^+$ denotes the set of positive integers.)

EXAMPLE 3

Show that the inclusion relation $\subseteq$ is a partial ordering on the power set of a set S .

Solution: Because $A \subseteq A$ whenever A is a subset of S , $\subseteq$ is reflexive. It is antisymmetric because $A \subseteq B$ and $B \subseteq A$ imply that $A = B$. Finally, $\subseteq$ is transitive, because $A \subseteq B$ and $B \subseteq C$ imply that $A \subseteq C$. Hence, $\subseteq$ is a partial ordering on $P(S)$, and $(P(S), \subseteq)$ is a poset.

EXAMPLE 4 Let R be the relation on the set of people such that xRy if x and y are people and x is older than y . Show that R is not a partial ordering.



Solution: Note that R is antisymmetric because if a person x is older than a person y , then y is not older than x . That is, if xRy , then $y \not R x$. The relation R is transitive because if person x is older than person y and y is older than person z , then x is older than z . That is, if xRy and yRz , then xRz . However, R is not reflexive, because no person is older than himself or herself. That is, $x \not R x$ for all people x . It follows that R is not a partial ordering. 



Def 2.

The elements a and b of a poset $(S, \preceq)$ are called **comparable** if either $a \preceq b$ or $b \preceq a$. When a and b are elements of S such that neither $a \preceq b$ or $b \preceq a$, a and b are called **incomparable**.

Example 5.

In the poset $(\mathbf{Z}^+, |)$, are the integers 3 and 9 comparable? Are 5 and 7 comparable?

Sol :

$$3|9 \Rightarrow \text{comparable}$$

$$5 \nmid 7 \text{ and } 7 \nmid 5 \Rightarrow \text{incomparable}$$



Def 3. If $(S, \preceq)$ is a poset and every two elements of S are comparable, S is called a **totally ordered** or **linearly ordered set**, and $\preceq$ is called a **total order** (全序) or a **linear order**. A totally ordered set is also called a **chain**.

Example 6.

The poset $(\mathbf{Z}, \leq)$ is totally ordered, because $a \leq b$ or $b \leq a$ whenever a and b are integers.

Example 7.

The poset $(\mathbf{Z}^+, |)$ is not totally ordered.



DEFINITION 4

$(S, \preccurlyeq)$ is a *well-ordered set* if it is a poset such that $\preccurlyeq$ is a total ordering and every nonempty subset of S has a least element.

EXAMPLE 8

The set of ordered pairs of positive integers, $\mathbf{Z}^+ \times \mathbf{Z}^+$, with $(a_1, a_2) \preccurlyeq (b_1, b_2)$ if $a_1 < b_1$, or if $a_1 = b_1$ and $a_2 \leq b_2$ (the lexicographic ordering), is a well-ordered set. The verification of this is left as an exercise at the end of this section. The set $\mathbf{Z}$, with the usual $\leq$ ordering, is not well-ordered because the set of negative integers, which is a subset of $\mathbf{Z}$, has no least element. 

■ Lexicographic Order (字典順序)

less than (in A_2) the second entry of the second pair. In other words, (a_1, a_2) is less than (b_1, b_2) , that is,

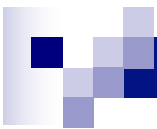
$$(a_1, a_2) < (b_1, b_2),$$

either if $a_1 <_1 b_1$ or if both $a_1 = b_1$ and $a_2 <_2 b_2$.

EXAMPLE 9 Determine whether $(3, 5) < (4, 8)$, whether $(3, 8) < (4, 5)$, and whether $(4, 9) < (4, 11)$ in the poset $(\mathbf{Z} \times \mathbf{Z}, \preceq)$, where $\preceq$ is the lexicographic ordering constructed from the usual $\leq$ relation on $\mathbf{Z}$.

Solution: Because $3 < 4$, it follows that $(3, 5) < (4, 8)$ and that $(3, 8) < (4, 5)$. We have $(4, 9) < (4, 11)$, because the first entries of $(4, 9)$ and $(4, 11)$ are the same but $9 < 11$. 

EXAMPLE 10 Note that $(1, 2, 3, 5) < (1, 2, 4, 3)$, because the entries in the first two positions of these 4-tuples agree, but in the third position the entry in the first 4-tuple, 3, is less than that in the second 4-tuple, 4. (Here the ordering on 4-tuples is the lexicographic ordering that comes from the usual “less than or equals” relation on the set of integers.) 



Example 9.

In the poset $(\mathbf{Z} \times \mathbf{Z}, \preceq)$, where $\preceq$ is the lexicographic ordering constructed from the usual $\leq$ relation on $\mathbf{Z}$.

$$(3, 5) \prec (4, 8),$$

$$(3, 8) \prec (4, 5),$$

$$(4, 9) \prec (4, 11)$$

Hasse Diagrams (用來描述poset中元素的大小關係)

將poset用圖形表示，若 a, b 是comparable且 $a \prec b$ ，則將 ab 間連一條邊，並將 b 節點放在 a 節點的上邊，但若 $a \prec b \prec c$ ，則不畫 ac 間的邊。

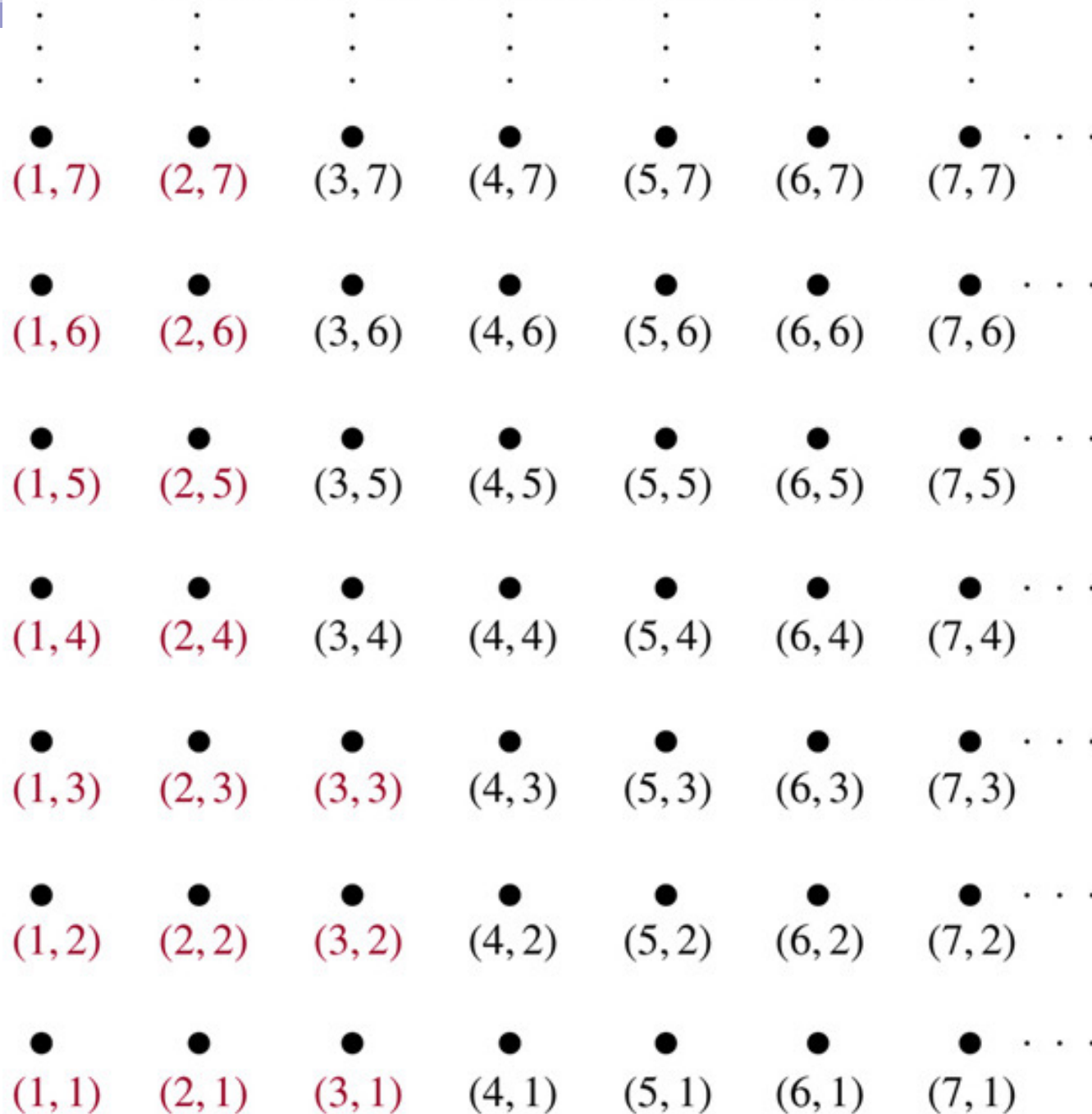


FIGURE 1

The Ordered
Pairs Less Than
 $(3, 4)$ in
Lexicographic
Order.



EXAMPLE 11 Consider the set of strings of lowercase English letters. Using the ordering of letters in the alphabet, a lexicographic ordering on the set of strings can be constructed. A string is less than a second string if the letter in the first string in the first position where the strings differ comes before the letter in the second string in this position, or if the first string and the second string agree in all positions, but the second string has more letters. This ordering is the same as that used in dictionaries. For example,

discreet $\prec$ *discrete*,

because these strings differ first in the seventh position, and $e \prec t$. Also,

discreet $\prec$ *discreetness*,

because the first eight letters agree, but the second string is longer. Furthermore,

discrete $\prec$ *discretion*,

because

discrete $\prec$ *discreti*.



■ Hasse Diagrams (哈斯圖)

Many edges in the directed graph for a finite poset do not have to be shown because they must be present. For instance, consider the directed graph for the partial ordering $\{(a, b) \mid a \leq b\}$ on the set $\{1, 2, 3, 4\}$, shown in Figure 2(a). Because this relation is a partial ordering, it is reflexive, and its directed graph has loops at all vertices. Consequently, we do not have to show these loops because they must be present; in Figure 2(b) loops are not shown. Because a partial ordering is transitive, we do not have to show those edges that must be present because of transitivity. For example, in Figure 2(c) the edges $(1, 3)$, $(1, 4)$, and $(2, 4)$ are not shown because they must be present. If we assume that all edges are pointed “upward” (as they are drawn in the figure), we do not have to show the directions of the edges; Figure 2(c) does not show directions.

In general, we can represent a partial ordering on a finite set using this procedure: Start with the directed graph for this relation. Because a partial ordering is reflexive, a loop is present at every vertex. Remove these loops. Remove all edges that must be in the partial ordering because of the presence of other edges and transitivity. For instance, if (a, b) and (b, c) are in the partial ordering, remove the edge (a, c) , because it must be present also. Furthermore, if (c, d) is also in the partial ordering, remove the edge (a, d) , because it must be present also. Finally, arrange each edge so that its initial vertex is below its terminal vertex (as it is drawn on paper). Remove all the arrows on the directed edges, because all edges point “upward” toward their terminal vertex. (The edges left correspond to pairs in the covering relation of the poset. See the preamble to Exercise 28.)

These steps are well defined, and only a finite number of steps need to be carried out for a finite poset. When all the steps have been taken, the resulting diagram contains sufficient information to find the partial ordering. This diagram is called a **Hasse diagram**, named after the twentieth-century German mathematician Helmut Hasse.

■ Hasse Diagrams (哈斯圖)

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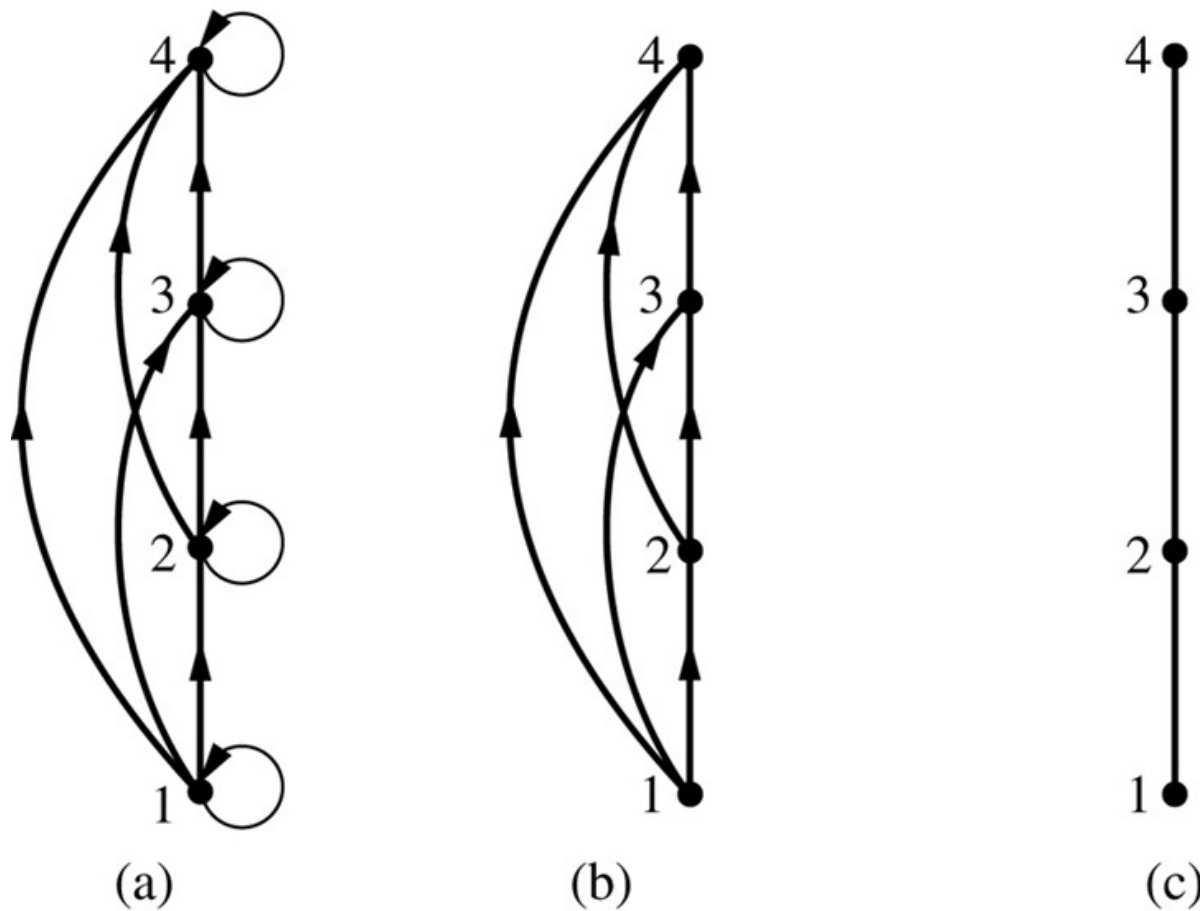


FIGURE 2 Constructing the Hasse Diagram for $(\{1, 2, 3, 4\}, \leq)$.

EXAMPLE 12 Draw the Hasse diagram representing the partial ordering $\{(a, b) | a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$.

Solution: Begin with the digraph for this partial order, as shown in Figure 3(a). Remove all loops, as shown in Figure 3(b). Then delete all the edges implied by the transitive property. These are $(1, 4)$, $(1, 6)$, $(1, 8)$, $(1, 12)$, $(2, 8)$, $(2, 12)$, and $(3, 12)$. Arrange all edges to point upward, and delete all arrows to obtain the Hasse diagram. The resulting Hasse diagram is shown in Figure 3(c).

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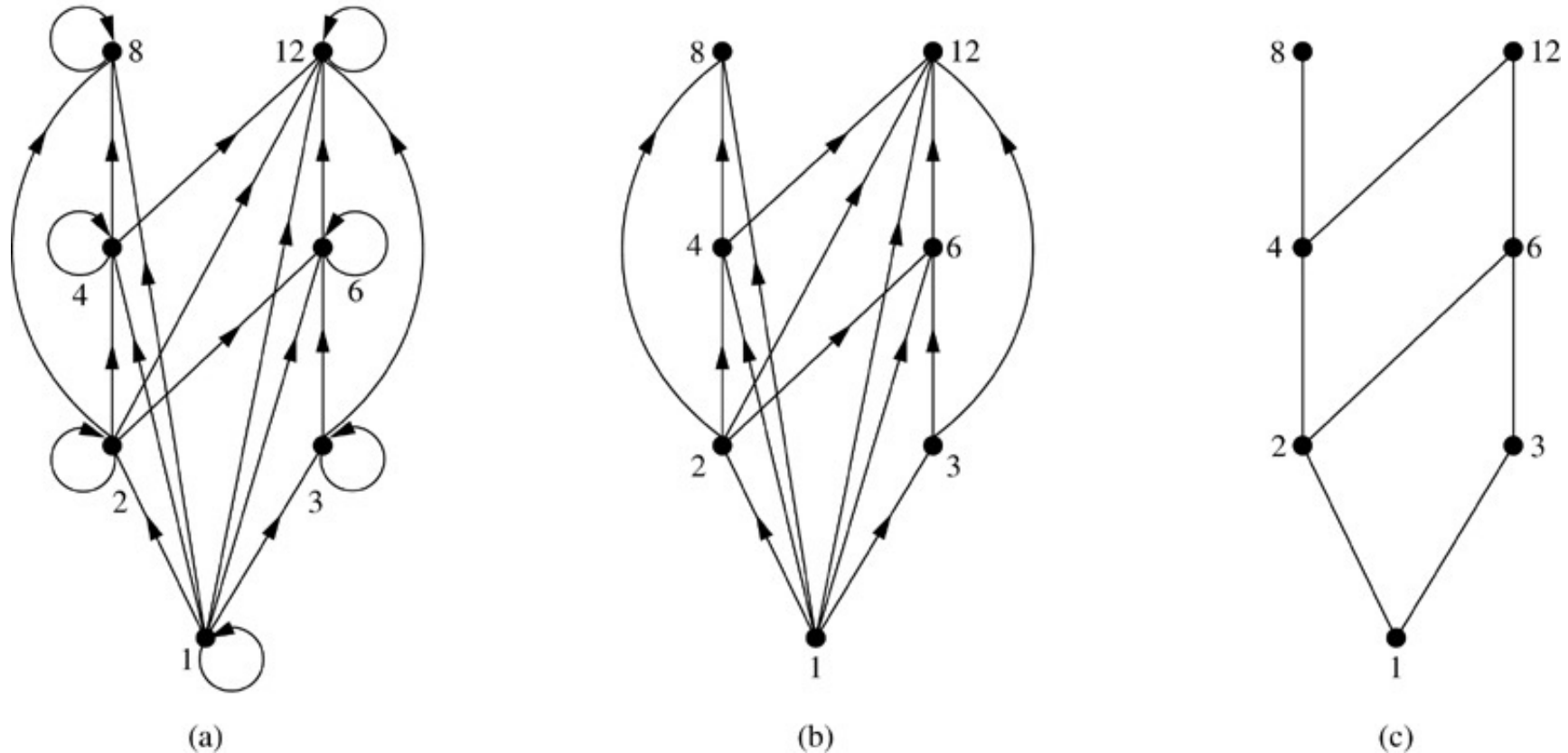


FIGURE 3 Constructing the Hasse Diagram of $(\{1, 2, 3, 4, 6, 8, 12\}, |)$.

EXAMPLE 13 Draw the Hasse diagram for the partial ordering $\{(A, B) \mid A \subseteq B\}$ on the power set $P(S)$ where $S = \{a, b, c\}$.

Solution: The Hasse diagram for this partial ordering is obtained from the associated digraph by deleting all the loops and all the edges that occur from transitivity, namely, $(\emptyset, \{a, b\})$, $(\emptyset, \{a, c\})$, $(\emptyset, \{b, c\})$, $(\emptyset, \{a, b, c\})$, $(\{a\}, \{a, b, c\})$, $(\{b\}, \{a, b, c\})$, and $(\{c\}, \{a, b, c\})$. Finally all edges point upward, and arrows are deleted. The resulting Hasse diagram is illustrated in Figure 4.

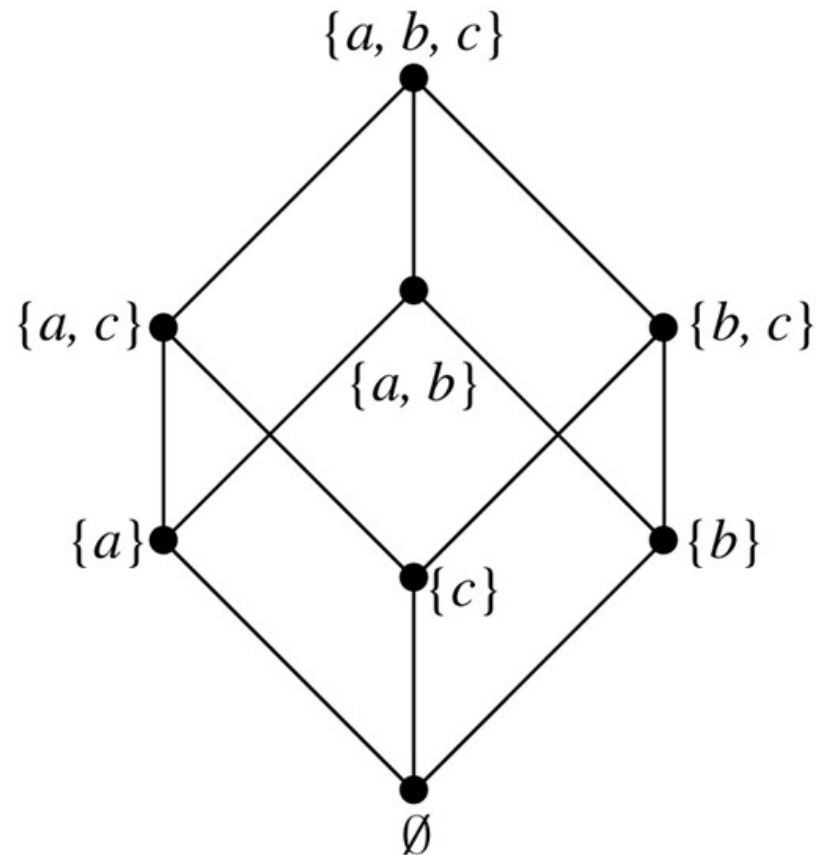


FIGURE 4 The Hasse Diagram of $(P(\{a, b, c\}), \subseteq)$.

EXAMPLE 14 Which elements of the poset $(\{2, 4, 5, 10, 12, 20, 25\}, |)$ are maximal, and which are minimal?

Solution: The Hasse diagram in Figure 5 for this poset shows that the maximal elements are 12, 20, and 25, and the minimal elements are 2 and 5. As this example shows, a poset can have more than one maximal element and more than one minimal element. 

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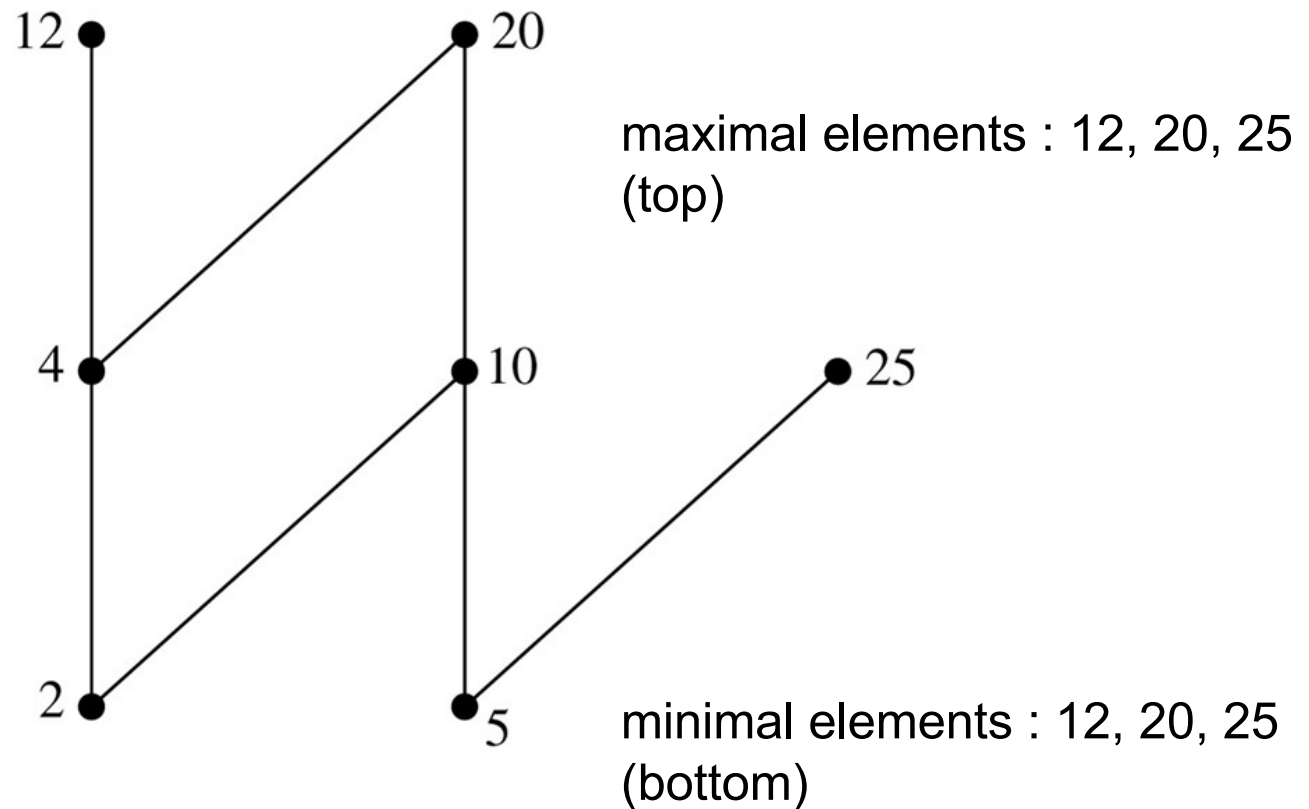


FIGURE 5 The Hasse Diagram of a Poset.



Sometimes there is an element in a poset that is greater than every other element. Such an element is called the greatest element. That is, a is the **greatest element** of the poset $(S, \preceq)$ if $b \preceq a$ for all $b \in S$. The greatest element is unique when it exists [see Exercise 40(a) at the end of this section]. Likewise, an element is called the least element if it is less than all the other elements in the poset. That is, a is the **least element** of $(S, \preceq)$ if $a \preceq b$ for all $b \in S$. The least element is unique when it exists [see Exercise 40(b) at the end of the section].

Maximal and Minimal Elements

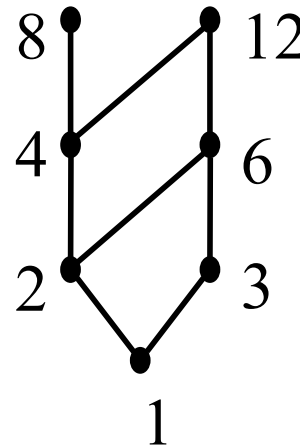
Def.

An element $a \in S$ is **maximal** in the poset $(S, \preceq)$ if there is no $b \in S$ such that $a \prec b$. Similarly, an element $a \in S$ is **minimal** if there is no $b \in S$ such that $b \prec a$.

a is the **greatest element** of the poset $(S, \preceq)$ if $b \preceq a$ for all $b \in S$. a is the **least element** of $(S, \preceq)$ if $a \preceq b$ for all $b \in S$.

Example 12 中

8, 12 是 maximal ,
1 是 least 也是 minimal ,
沒有 greatest element



EXAMPLE 15

Determine whether the posets represented by each of the Hasse diagrams in Figure 6 have a greatest element and a least element.

Solution: The least element of the poset with Hasse diagram (a) is a . This poset has no greatest element. The poset with Hasse diagram (b) has neither a least nor a greatest element. The poset with Hasse diagram (c) has no least element. Its greatest element is d . The poset with Hasse diagram (d) has least element a and greatest element d . 

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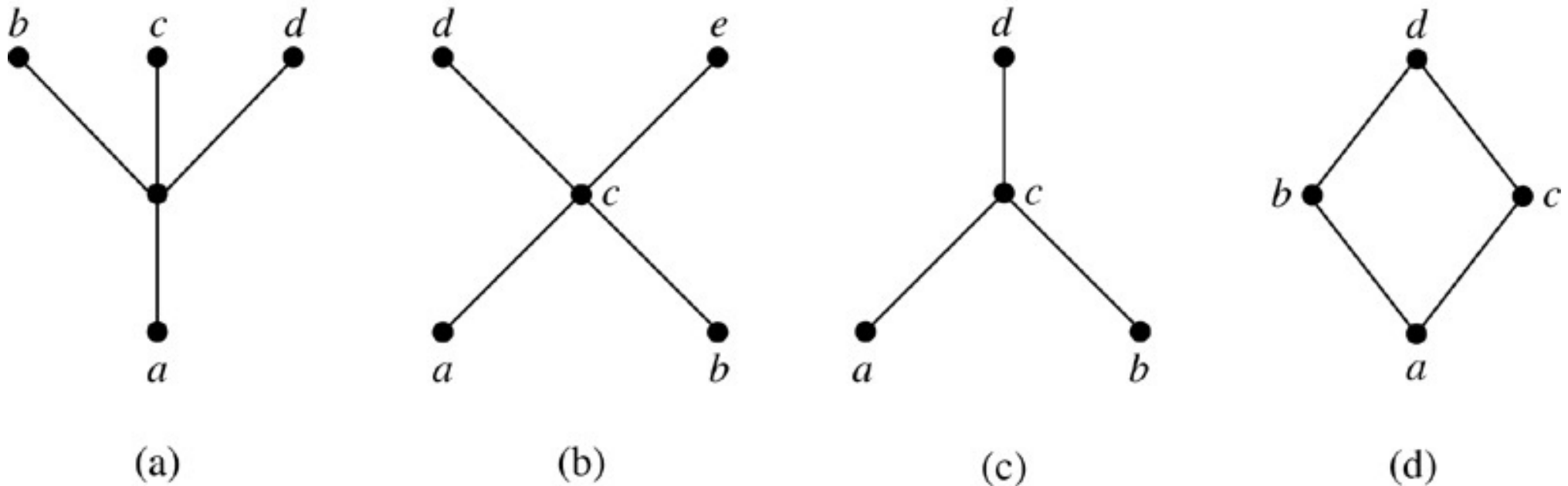


FIGURE 6 Hasse Diagrams of Four Posets.

EXAMPLE 16 Let S be a set. Determine whether there is a greatest element and a least element in the poset $(P(S), \subseteq)$.

Solution: The least element is the empty set, because $\emptyset \subseteq T$ for any subset T of S . The set S is the greatest element in this poset, because $T \subseteq S$ whenever T is a subset of S . ▶

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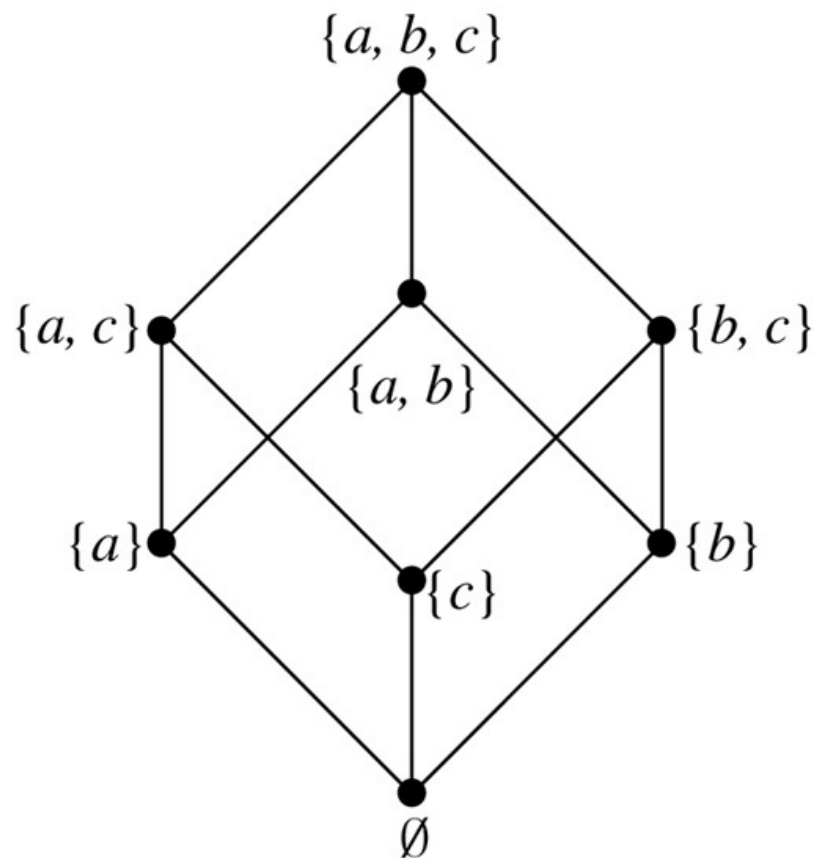
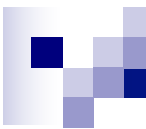


FIGURE 4 The Hasse Diagram of $(P(\{a, b, c\}), \subseteq)$.



Sometimes it is possible to find an element that is greater than or equal to all the elements in a subset A of a poset $(S, \preceq)$. If u is an element of S such that $a \preceq u$ for all elements $a \in A$, then u is called an **upper bound** of A . Likewise, there may be an element less than or equal to all the elements in A . If l is an element of S such that $l \preceq a$ for all elements $a \in A$, then l is called a **lower bound** of A .

EXAMPLE 20 Find the greatest lower bound and the least upper bound of the sets $\{3, 9, 12\}$ and $\{1, 2, 4, 5, 10\}$, if they exist, in the poset $(\mathbf{Z}^+, |)$.



Solution: An integer is a lower bound of $\{3, 9, 12\}$ if 3, 9, and 12 are divisible by this integer. The only such integers are 1 and 3. Because $1 \mid 3$, 3 is the greatest lower bound of $\{3, 9, 12\}$. The only lower bound for the set $\{1, 2, 4, 5, 10\}$ with respect to $|$ is the element 1. Hence, 1 is the greatest lower bound for $\{1, 2, 4, 5, 10\}$.

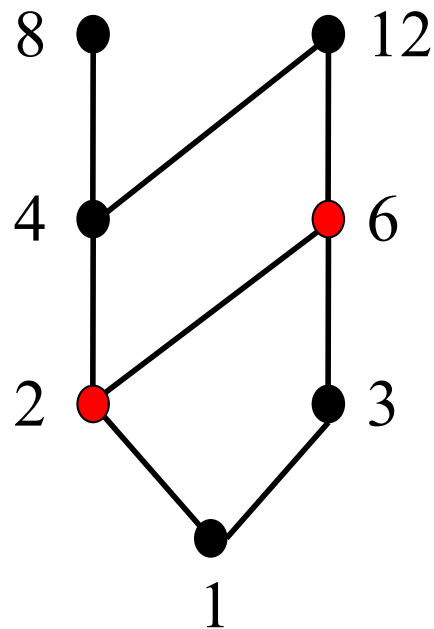
An integer is an upper bound for $\{3, 9, 12\}$ if and only if it is divisible by 3, 9, and 12. The integers with this property are those divisible by the least common multiple of 3, 9, and 12, which is 36. Hence, 36 is the least upper bound of $\{3, 9, 12\}$. A positive integer is an upper bound for the set $\{1, 2, 4, 5, 10\}$ if and only if it is divisible by 1, 2, 4, 5, and 10. The integers with this property are those integers divisible by the least common multiple of these integers, which is 20. Hence, 20 is the least upper bound of $\{1, 2, 4, 5, 10\}$. 

Def.

Let A be a subset of a poset $(S, \preceq)$. If u is an element of S such that $a \preceq u$ for all elements $a \in A$, then u is called an **upper bound** of A .

If l is an element of S such that $l \preceq a$ for all elements $a \in A$, then l is called an **lower bound** of A .

Ex



$$A = \{2, 6\}$$

upper bound of A : 6, 12

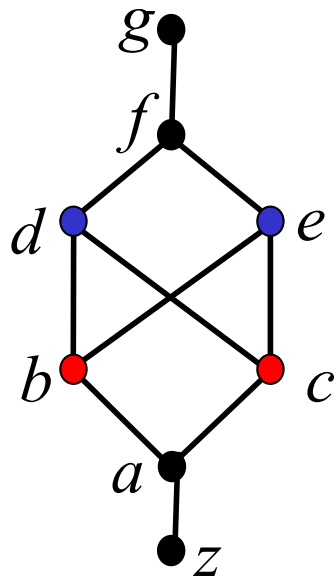
lower bound of A : 1, 2

Def.

Let A be a subset of a poset $(S, \preceq)$. An element x is called the **least upper bound** of A if x is an upper bound of A and $x \preceq z$ whenever z is an upper bound of A .

Let A be a subset of a poset $(S, \preceq)$. An element x is called the **greatest lower bound** of A if x is a lower bound of A and $y \preceq x$ whenever y is a lower bound of A .

Ex



$$A_1 = \{d, e\}, A_2 = \{b, c\}$$

least upper bound of $A_1 = f$

A_1 has no greatest lower bound

A_2 has no least upper bound

greatest lower bound of $A_2 = a$

EXAMPLE 18 Find the lower and upper bounds of the subsets $\{a, b, c\}$, $\{j, h\}$, and $\{a, c, d, f\}$ in the poset with the Hasse diagram shown in Figure 7.

Solution: The upper bounds of $\{a, b, c\}$ are e, f, j , and h , and its only lower bound is a . There are no upper bounds of $\{j, h\}$, and its lower bounds are a, b, c, d, e , and f . The upper bounds of $\{a, c, d, f\}$ are f, h , and j , and its lower bound is a . ◀

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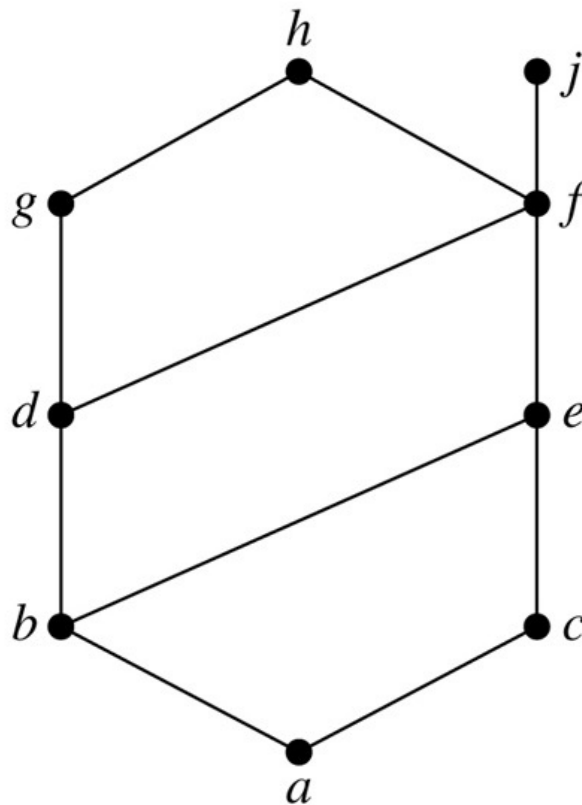
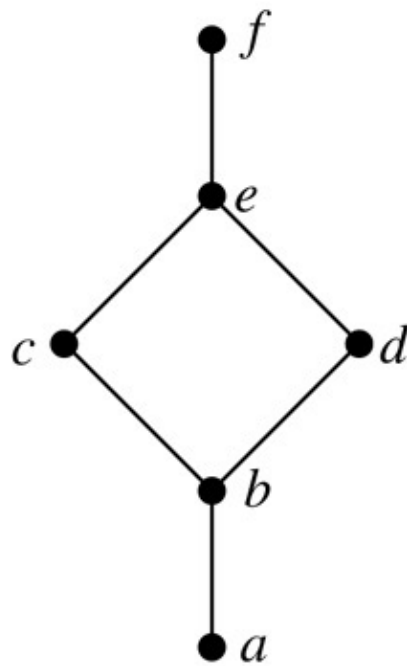


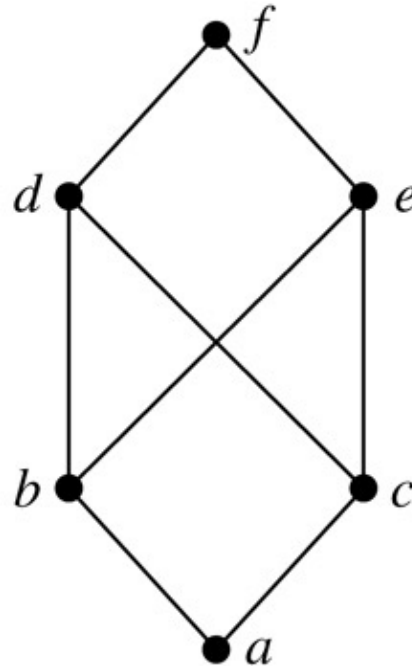
FIGURE 7 The Hasse Diagram of a Poset.

A partially ordered set in which every pair of elements has both a least upper bound and a greatest lower bound is called a **lattice**. Lattices have many special properties. Furthermore, lattices are used in many different applications such as models of information flow and play an important role in Boolean algebra.

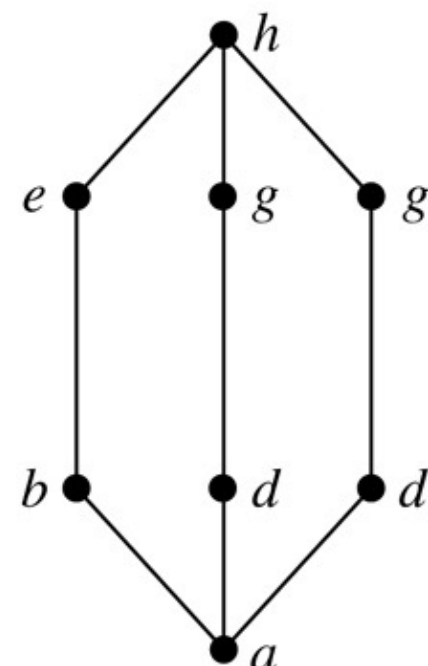
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(a)



(b)



(c)

FIGURE 8 Hasse Diagrams of Three Posets.



EXAMPLE 22 Is the poset $(\mathbb{Z}^+, |)$ a lattice?

Solution: Let a and b be two positive integers. The least upper bound and greatest lower bound of these two integers are the least common multiple and the greatest common divisor of these integers, respectively, as the reader should verify. It follows that this poset is a lattice. 

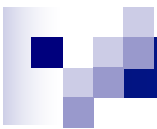
EXAMPLE 23 Determine whether the posets $(\{1, 2, 3, 4, 5\}, |)$ and $(\{1, 2, 4, 8, 16\}, |)$ are lattices.

Solution: Because 2 and 3 have no upper bounds in $(\{1, 2, 3, 4, 5\}, |)$, they certainly do not have a least upper bound. Hence, the first poset is not a lattice.

Every two elements of the second poset have both a least upper bound and a greatest lower bound. The least upper bound of two elements in this poset is the larger of the elements and the greatest lower bound of two elements is the smaller of the elements, as the reader should verify. Hence, this second poset is a lattice. 

EXAMPLE 24 Determine whether $(P(S), \subseteq)$ is a lattice where S is a set.

Solution: Let A and B be two subsets of S . The least upper bound and the greatest lower bound of A and B are $A \cup B$ and $A \cap B$, respectively, as the reader can show. Hence, $(P(S), \subseteq)$ is a lattice. 



Example 22.

Is the poset $(\mathbf{Z}^+, |)$ a lattice?

Sol : For any $a, b \in \mathbf{Z}^+$,
 $\gcd(a, b)$ is the greatest lower bound of a, b .
 $\text{lcm}(a, b)$ is the least upper bound of a, b .
 $\Rightarrow$ Yes

Example 23.

Determine whether the posets $(\{1, 2, 3, 4, 5\}, |)$ and $(\{1, 2, 4, 8, 16\}, |)$ are lattices.

Sol : In $(\{1, 2, 3, 4, 5\}, |)$, 2 and 3 has no l.u.b. $\Rightarrow$ No.

In $(\{1, 2, 4, 8, 16\}, |)$,
any a, b has l.u.b. and g.l.b. $\Rightarrow$ Yes.



Topological Sorting

Suppose that a project is made up of 20 different tasks. Some tasks can be completed only after others have been finished. How can an order be found for these tasks?

Def.

A total ordering $\preceq$ is said to be **compatible** (相容) with the partial ordering R if $a \preceq b$ whenever aRb . Constructing a compatible total ordering from a partial ordering is called **topological sorting**.

Lemma 1.

Every finite nonempty poset $(S, \preceq)$ has at least one minimal element.

Topological Sorting

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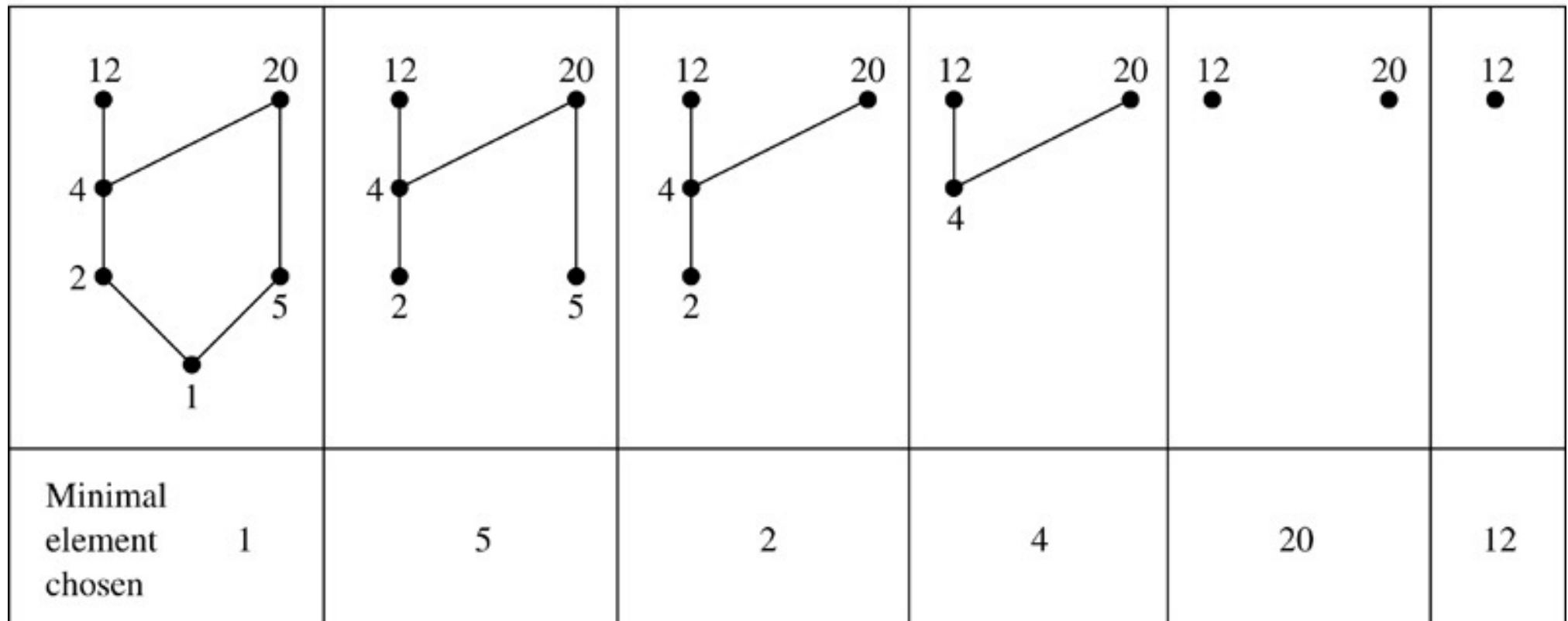


FIGURE 9 A Topological Sort of $(\{1, 2, 4, 5, 12, 20\}, |)$.

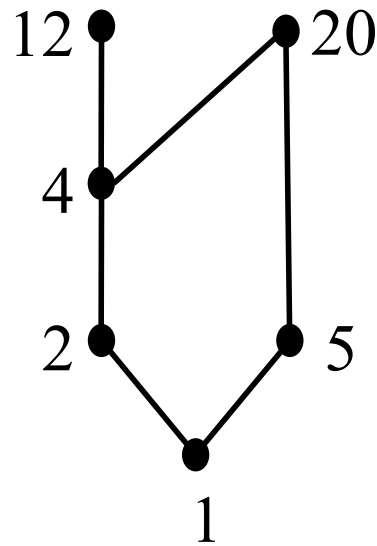
EXAMPLE 26 Find a compatible total ordering for the poset $(\{1, 2, 4, 5, 12, 20\}, |)$.

Topological sorting 的方式： 逐次output minimal element，
即得到由小到大的compatible total ordering

Example 26.

Find a compatible total ordering for the poset $(\{1, 2, 4, 5, 12, 20\}, |)$.

Sol :



$1 \prec 2 \prec 5 \prec 4 \prec 12 \prec 20$

2跟5的顺序可交换，12跟20也是

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FIG

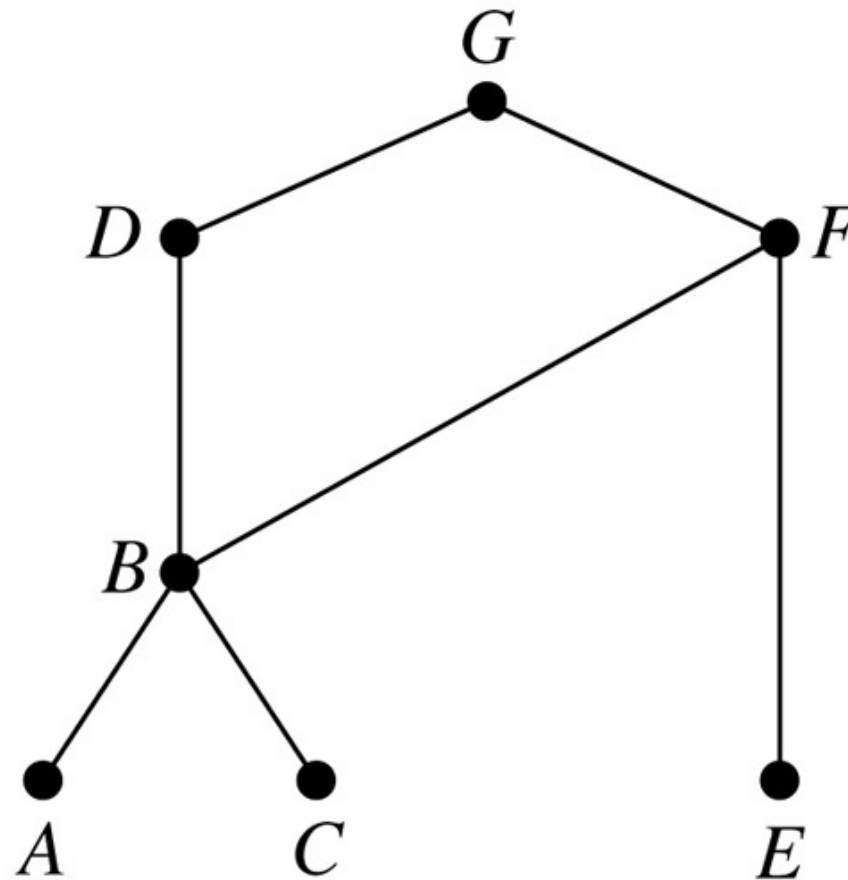


FIGURE 10 The Hasse Diagram for Seven Tasks.

FIGURE 11 (9.6)

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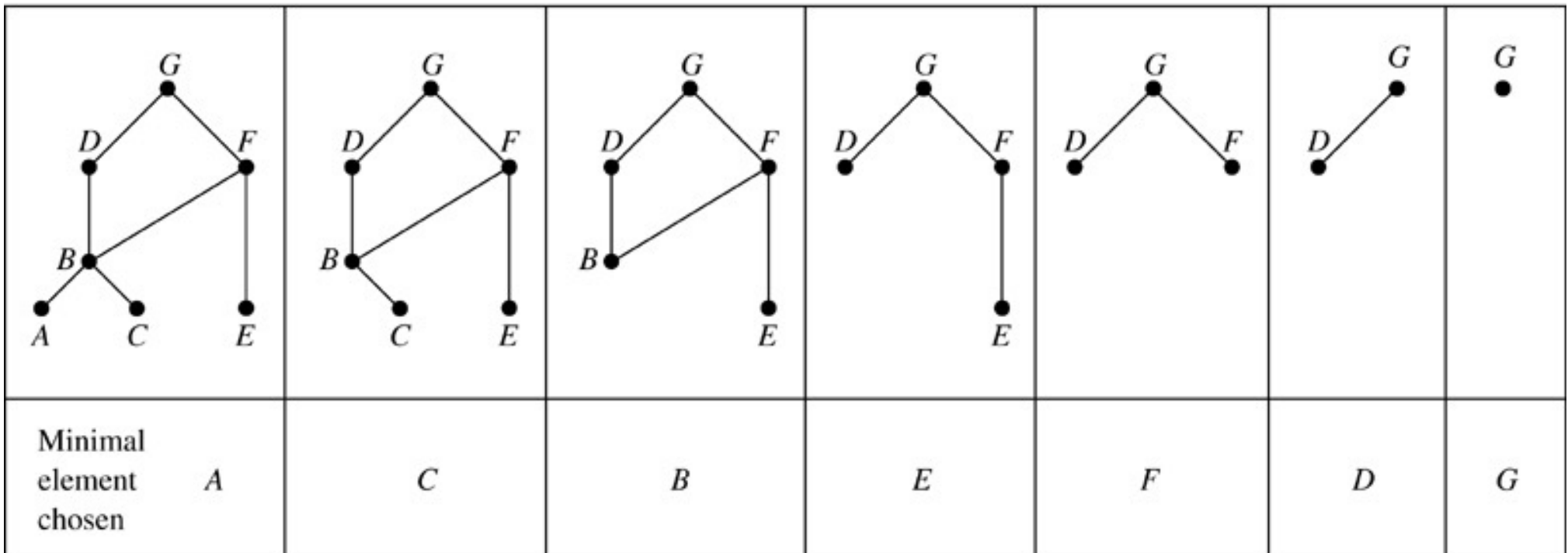


FIGURE 11 A Topological Sort of the Tasks.



7. Determine whether the relations represented by these zero-one matrices are partial orders.

a)
$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b)
$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

c)
$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$



14. Which of these pairs of elements are comparable in the poset $(\mathbf{Z}^+, |)$?

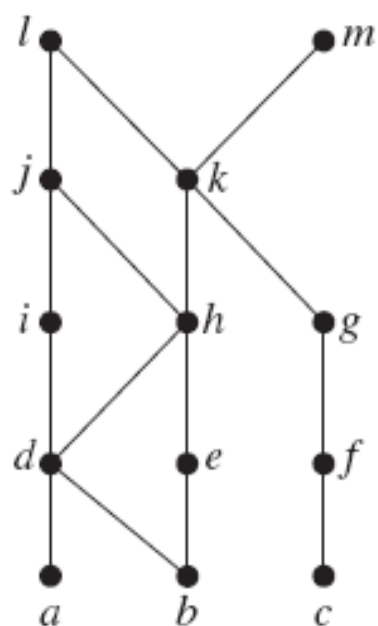
- a)** 5, 15 **b)** 6, 9 **c)** 8, 16 **d)** 7, 7

18. Find the lexicographic ordering of these strings of lower-case English letters:

- a)** *quack, quick, quicksilver, quicksand, quacking*
b) *open, opener, opera, operand, opened*
c) *zoo, zero, zoom, zoology, zoological*



32. Answer these questions for the partial order represented by this Hasse diagram.



- a) Find the maximal elements.
- b) Find the minimal elements.
- c) Is there a greatest element?
- d) Is there a least element?
- e) Find all upper bounds of $\{a, b, c\}$.
- f) Find the least upper bound of $\{a, b, c\}$, if it exists.
- g) Find all lower bounds of $\{f, g, h\}$.
- h) Find the greatest lower bound of $\{f, g, h\}$, if it exists.